

END TERM EXAMINATION [NOV-DEC 2018]

THIRD SEMESTER [B.TECH]

ANALOG ELECTRONICS-I [ETEC-203]

Time : 3 hrs.

M.M. : 75

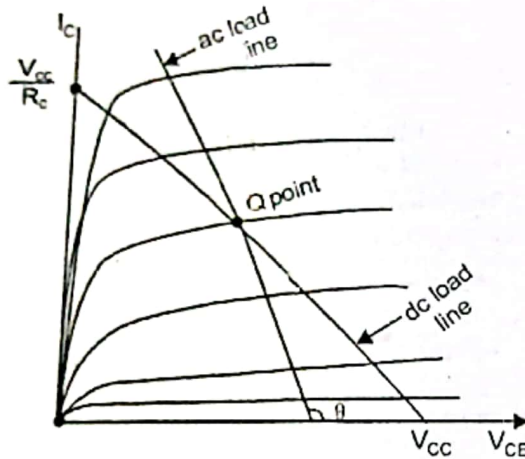
Note: Attempt five questions in all including Q. No. 1 which is compulsory.

Q. 1. (a) Why do we have A.C and D.C. load lines? (2.5)

Ans. AC load line: This is the line on the output characteristics of a transistor which gives the value of V_{CE} and I_C when an ac signal is applied. The slope of the ac load

line is given as $\frac{-1}{R_{AC}}$ where $R_{AC} = R_C \parallel R_L$ and represents the ac load resistance.

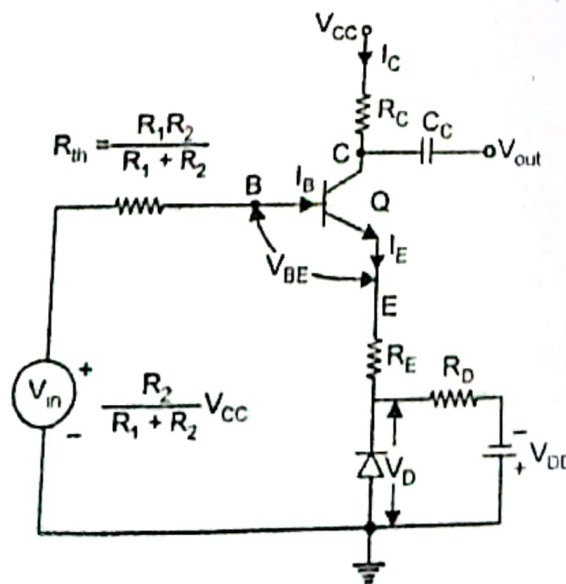
D.C load line: This is the line on the output characteristics of a transistor which helps to determine the operating point of transistor when only dc supply is applied. For faithful amplification of input signal a judicious selection of operating or Q-point is of utmost importance. The Q point must be established in the centre of the dc load line.



$$\theta = \tan^{-1}\left(-\frac{1}{R_{AC}}\right)$$

Q. 1. (b) Explain bias compensation with the help of diode. (2.5)

Ans. Diode Compensation for variation in V_{BE} .



Thevenin's equivalent circuit of self bias stabilization and diode compensation is shown

By KVL

$$V_{th} = I_b R_{th} + V_{BE} + I_E R_E - V_D = I_B R_{th} + V_{BE} + (I_B + I_C) R_E - V_D$$

$$= I_B (R_{th} + R_E) + I_C R_E + V_{BE} - V_D$$

$$V_{th} = \left[\frac{I_C - (1 + \beta) I_{CO}}{\beta} \right] (R_{th} + R_E) + I_C R_E + V_{BE} - V_D$$

or

$$I_C = [R_{th} + (1 + \beta) R_E] = \beta [V_{th} - (V_{BE} - V_D)] + (1 + \beta) I_{CO} (R_{th} + R_E)$$

$$I_C = \frac{\beta [V_{th} - (V_{BE} - V_D)] + (1 + \beta) I_{CO} (R_{th} + R_E)}{R_{th} + (1 + \beta) R_E}$$

Since variation in V_{BE} and V_D are the same due to temperature variation, so $(V_{BE} - V_D)$ remains unchanged and I_C becomes insensitive to variation in V_{BE} .

Diode Compensation for I_{CO} : The diode and Transistor used in the circuit are of the same material. So I_{CO} of transistor and reverse saturation current I_0 of diode will increase at the same rate

$$I = \frac{V_{CC} - V_{BE}}{R_1} \approx \frac{V_{CC}}{R_1} \quad \dots(1)$$

($\because V_{BE} = 0.3$ for Ge)

Now

$$I_B = I - I_0$$

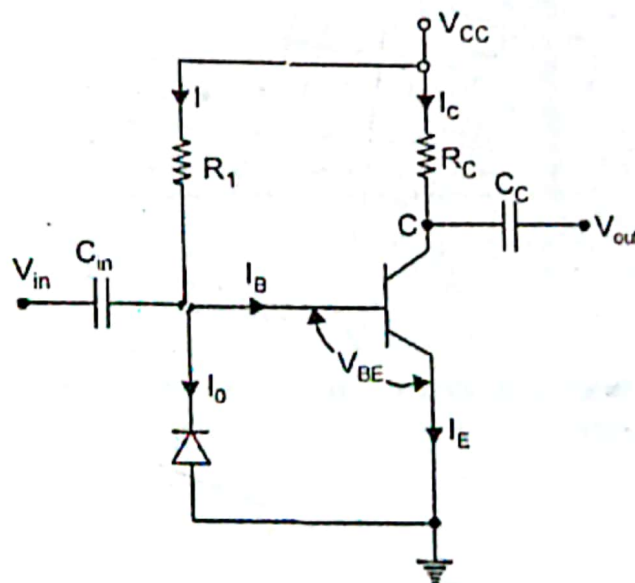
and

$$I_C = \beta I_B + (1 + \beta) I_{CO} \quad \dots(2)$$

or

$$I_C = \beta I - \beta I_0 + (1 + \beta) I_{CO} \quad \dots(3)$$

If $\beta \gg 1$ and I_0 & I_{CO} track each other over the desired temperature range, then collector current I_C remains constant.



Q. 1. (c) What are the drawbacks of using multistage amplifiers? (2.5)

Ans. Drawbacks of multistage amplifiers:

→ The overall gain of multistage amplifier is less than the product of the individual stages due to the loading effect.

→ Multistage amplifier is more complicated

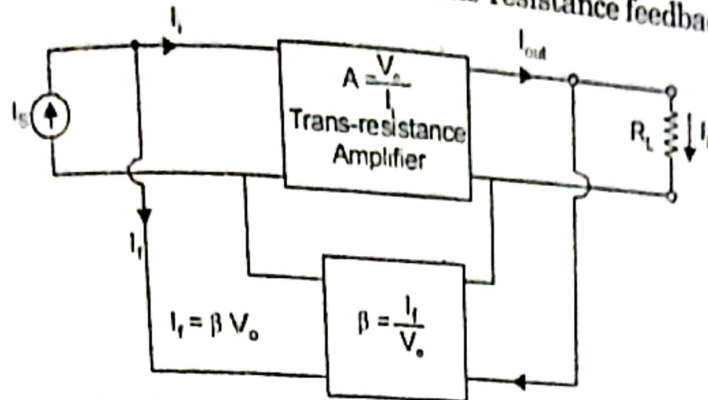
→ It requires more chip area because capacitors will occupy large area on chip.

→ Propagation delay increases. (2.5)

Q. 1. (d) How can the gain of darlington pair be calculated?

Ans. Refer Q. 1. (c) First Term Examination 2018.

Q. 1. (e) Draw the block diagram of trans resistance feedback configuration.
 Ans. Voltage shunt feedback is also known as trans-resistance feedback configuration. (2.5)



Q. 1. (f) "Negative feedback is necessary for amplification". True or False and justify the statement? (2.5)

Ans. It is true that negative feedback is necessary for the amplification because it improves its performance in several aspects and does not effect the current of the circuit.

Advantages of negative feedback:

1. It improves the gain stability
2. It reduces non-linear distortion and noise.
3. It increases bandwidth or improved frequency response.
4. It increases input impedance.
5. It reduces output impedance.

Q. 1. (g) Why FET and MOSFET are required when BJT is there? (2.5)

Ans. The BJT is bipolar because the transistor operates with both types of charge carriers. Holes and electrons. The FET and MOSFET construction do not have a PN junction in their main current carrying path (channel), which can be made from N-type or P-type semiconductor material with high resistivity. The input impedance of FET and MOSFET is very high that is helpful to design a good amplifier. (2.5)

Q. 1. (h) How can we remove crossover distortion? (2.5)

Ans. Refer to Q. 9. (a) End Term Examination 2017.

Q. 1. (i) What is a Diac? Explain.

Ans. DIAC: DIAC is usually employed for triggering TRIAC. It is a two electrode bidirectional avalanche diode which can be switched from off-state to the on state for either polarity of the applied voltage. This just TRIAC without gate as shown in diagram operation. When applied voltage in either polarity is small (less voltage) a very small leakage current flows and the device remains in non-conducting mode. When the magnitude of the applied voltage exceeds the avalanche breakdown voltage, the DIAC current rises sharply as shown in characteristics.



Fig. Schematic Symbol

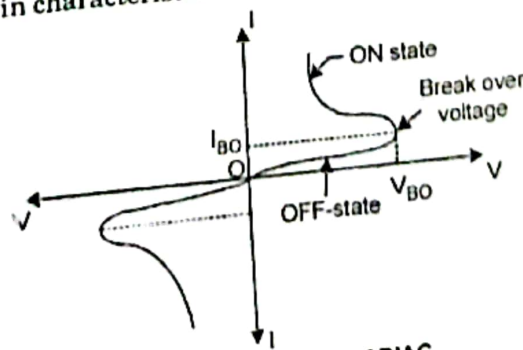


Fig. V-I characteristics of DIAC

Q. 1. (j) State and explain Barkhausen criterion. Also find the gain of an ideal oscillator? (2.5)

Ans. The use of positive feedback in amplifier having closed loop gain $|A_f|$ greater than 1 and satisfies the phase conditions will result in operation as an oscillator circuit.

The gain in positive feedback amplifier

$$A_f = \frac{A}{1 - A\beta}$$

The condition $A\beta = 1$ is known as Barkhausen criterion for self sustained oscillations. if

$$A\beta = 1$$

then

$$A_f = \frac{A}{1-1} = \infty \text{ (ideal condition)}$$

Q. 2. (a) What is thermal stability? Derive a condition for the same. (6.5)

Ans. Stability Factors: Refer Q. 1. (b) of End Term Examination 2017.

Thermal Runaway: Refer Q. 1. (a) of First Term Examination 2017.

Q. 2. (b) Derive a result for S in case of self bias. Explain whether the value of S in case of self bias will be smaller or larger than fixed bias and why? (6)

Ans.

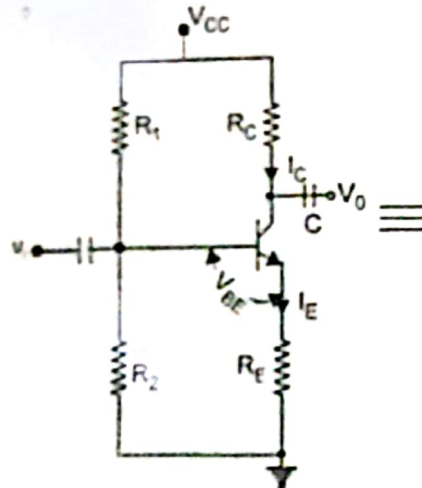


Fig. Self bias ckt

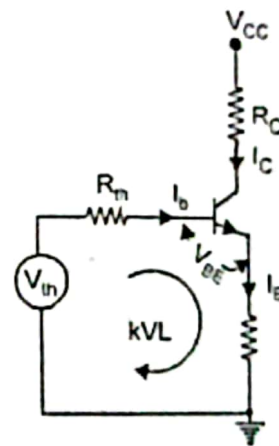


Fig. Thevenin's equivalent ckt

Thevenin's equivalent voltage $V_{th} = \frac{V_{CC} \times R_2}{R_1 + R_2}$

and resistance $R_{th} = R_1 \parallel R_2 = \frac{R_1 R_2}{R_1 + R_2}$

by KVL

$$V_{th} = I_B R_{th} + V_{BE} + I_E R_E$$

$$V_{th} = I_B R_{th} + V_{BE} + (I_B + I_C) R_E$$

differentiating w.r.t. I_C

$$0 = \frac{dI_B}{dI_C} (R_{th} + R_E) + R_E$$

$$\frac{dI_B}{dI_C} = \frac{-R_E}{(R_{th} + R_E)}$$

by using stability factor

$$S = \frac{1 + \beta}{1 - \beta \frac{dI_B}{dI_C}}$$

$$S = \frac{(1 + \beta)}{1 + \beta \frac{R_E}{(R_{Th} + R_E)}}$$

stability factor S in case of self bias will be smaller than fixed bias because the

base current I_B is independent of I_C α the value $\frac{dI_B}{dI_C} = 0$

$\therefore$ stability factor in case of fixed bias circuit

$S = (1 + \beta)$ which is greater than stability factor

$$S = \frac{(1 + \beta)}{1 + \beta \frac{R_E}{(R_{Th} + R_E)}} \text{ in case of self bias circuit.}$$

Q. 3. (a) Find $A_i, A_r/R_i, R_o$ for the C.E. configuration with by passed R_E . (6)

Ans. Refer to Q. 3. (a) First Term Examination 2017.

Q. 3. (b) Draw and explain Hybrid - model at high frequency and derive all parameters. (6.5)

Ans. Hybrid π or Giacoletto model which will be valid at high frequencies is indicated in the fig. (a) given below.

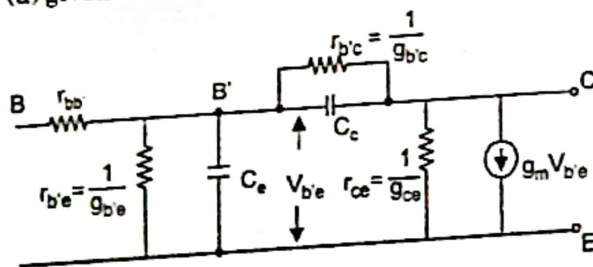


Fig. (a)

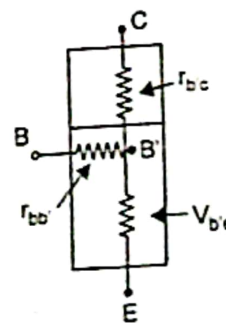


Fig. (b)

Fig. (b) is showing the ohmic base spreading resistance $r_{bb'}$ and virtual base B' . The Typical magnitudes for the elements of the hybrid- π model at room temperature and for $I_C = 1.3 \text{ mA}$ are

$$\begin{array}{llll} g_m = 50 \text{ mA/V} & r_{bb'} = 100 \Omega & r_{b'e} = 1 \text{ k} & r_{b'c} = 4 \text{ M} \\ r_{ce} = 80 \text{ K} & C_c = 3 \text{ PF} & C_e = 100 \text{ PF} & \end{array}$$

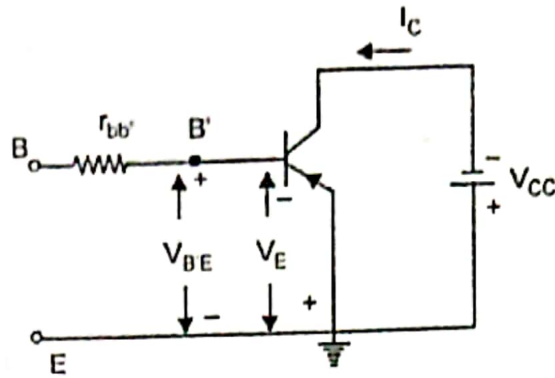
All the resistive components in the hybrid- π Model can be obtain from the h parameters in the CE configuration.

Transistor Transconductance (g_m)

In the active region the collector current is given with $\alpha_N = \alpha_0$

$$I_C = I_{C0} - \alpha_0 I_E$$

Transconductance g_m is defined

Fig. (c) Pertaining to the derivation of g_m .

by

$$g_m = \left. \frac{\partial I_C}{\partial V_{B'E}} \right|_{V_{CE}} = \alpha_0 \frac{\partial I_E}{\partial V_{B'E}} = \alpha_0 \frac{\partial I_E}{\partial V_E}$$

if emitter diode resistance

$$r_e = \frac{\partial V_E}{\partial I_E}, \text{ and hence}$$

$$g_m = \frac{\alpha_0}{r_e}$$

The dynamic resistance of a forward biased diode is given as $\frac{V_T}{I_E}$

$$\therefore g_m = \frac{\alpha_0 I_E}{V_T}$$

or

$$g_m \approx \frac{I_C}{V_T}$$

where $V_T = \frac{T}{11600}$. Note that g_m is directly proportional to current and inversely proportional to temperature.

$$g_m = \frac{|I_C|_{mA}}{26}$$

$\Rightarrow$ The input conductance ($g_{b'e}$): There is a hybrid- π model valid at low frequencies, where all capacitances are negligible

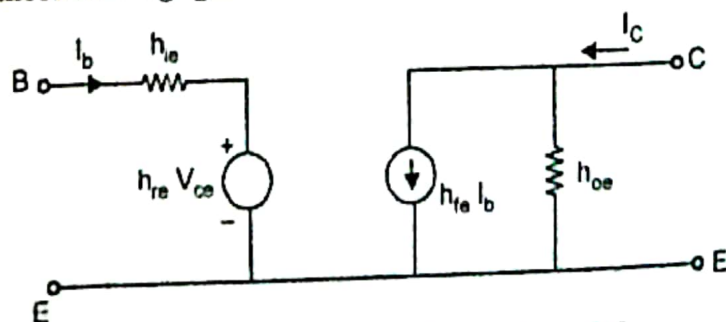


Fig. h-parameter equivalent model

$r_{b'e} \gg r_{b'e}$
 I_b flows into $r_{b'e}$ and $v_{b'e} = I_b r_{b'e}$. The short circuit current can be given by
 $I_C = g_m V_{b'e} \approx g_m I_b r_{b'e}$

and

$$h_{fe} = \left. \frac{I_c}{I_b} \right|_{V_{ce}} = g_m r_{b'e}$$

or

$$r_{b'e} = \frac{h_{fe}}{g_m} = \frac{h_{fe} V_T}{|I_C|}$$

$$g_{b'e} = \frac{g_m}{h_{fe}}$$

⇒ The feedback conductance ($g_{b'e}$): With open circuited, h_{re} is defined as the reverse voltage gain, with $I_b = 0$

$$\begin{aligned} h_{re} &= \frac{V_{b'e}}{V_{ce}} = \frac{r_{b'e}}{r_{b'e} + r_{b'c}} \\ r_{b'e}(1 - h_{re}) &= h_{re} r_{b'c} \\ r_{b'e} &= h_{re} r_{b'c} \quad (\because h_{re} \ll 1) \\ g_{b'e} &= h_{re} g_{b'c} \end{aligned}$$

or

⇒ The base spreading resistance ($r_{bb'}$): The input resistance with the output shorted is h_{ie} . under the conditions $r_{b'e}$ is in parallel with $r_{b'c}$ and $r_{b'e} \parallel r_{b'c} \approx r_{b'e}$

$$\begin{aligned} h_{ie} &= r_{bb'} + r_{b'e} \\ r_{bb'} &= h_{ie} - r_{b'e} \end{aligned}$$

and

$$h_{ic} = r_{bb'} + \frac{h_{fe} V_T}{|I_C|} \approx \frac{h_{fe} V_T}{|I_C|}$$

⇒ The output conductance (g_{ce}): with the input open circuited, this conductance is defined as h_{re} . For $I_b = 0$, we have

$$I_C = \frac{V_{ce}}{r_{ce}} + \frac{V_{ce}}{r_{b'e} + r_{b'c}} + g_m V_{b'e}$$

and

$$h_{oe} = \frac{I_c}{V_{ce}} = \frac{1}{r_{ce}} + \frac{1}{r_{b'e} + r_{b'c}} = g_m h_{re}$$

where we made use of the fact that $r_{b'c} \gg r_{b'e}$, we have

$$h_{oe} = g_{ce} + g_{b'c} + g_{b'e} h_{fe} \frac{g_{b'c}}{g_{b'e}}$$

$$g_{ce} = h_{oe} - (1 - h_{fe}) g_{b'c}$$

if

$$h_{fe} \gg 1 \text{ then}$$

$$g_{ce} \approx h_{oe} - g_m h_{re}$$

⇒ Hybrid- π Capacitances: Hybrid- π model includes two capacitances. The collector junction capacitance $C_C = C_{b'c}$ is the measured CB output capacitance with the input open ($I_E = 0$). Since in the active region the collector junction is reverse biased. Then C_C is the transition capacitance.

The capacitance C_e represents the sum of the emitter diffusion capacitance C_{De} and the emitter junction capacitance C_{Te} . For a forward biased emitter junction, C_{De} is usually much larger than, C_{Te} and hence

$$C_e = C_{De} + C_{Te} \approx C_{De}$$

Q. 4. (a) Compare cascaded and cascode amplifiers.

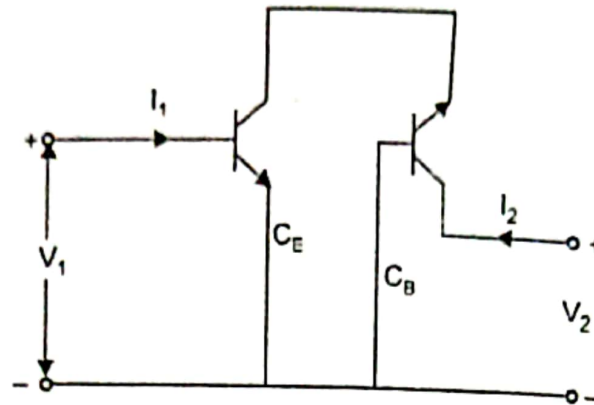
(6.5)

Ans. Cascode Amplifier: In a multistage amplifier, the output of first stage is combined to the next stage through a coupling device. This process is known as cascading to increase the overall voltage gain of amplifier.



Two stage cascaded amplifier

Cascode amplifiers: The CE-CB configuration is usually referred to as cascode amplifier. This configuration of course, has basically the input characteristics similar to those of CE amplifier and output characteristics similar to those of CB amplifier



Cascode pair

Q. 4. (b) Derive an expression for gain of multistage amplifier.

Ans. Refer to Q.1. (b) First Term Examination 2017.

Q. 5. (a) Derive gain with feedback, I/p and O/p resistances.

Ans. Series feedback connections tend to increase the input resistance while shunt feedback decrease the input impedance voltage feedback tends to decrease the output impedance while current feedback tends to increase the output impedance. Typically higher input and lower output impedances are desired for most cascade amplifiers. Both **there** are provided using the voltage-series feedback connection. Now we will examine the gain of each feedback connections.

Voltage-Series feedback

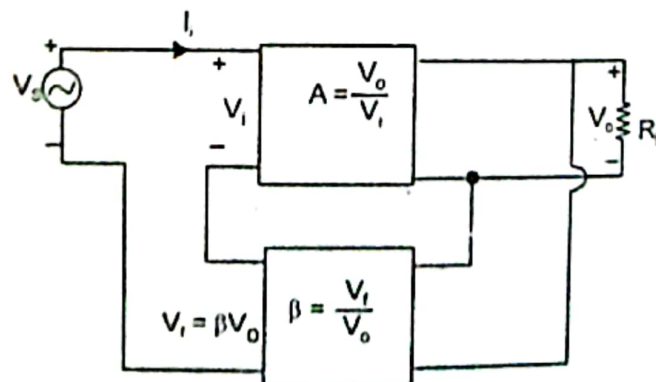


Fig. (a)

Gain without feedback $A = \frac{V_o}{V_i}$

if a feedback, V_f is connected in series with input, then

$$\begin{aligned} V_i &= V_s - V_f \\ V_o &= A V_i \\ &= A (V_s - V_f) = A V_s - A V_f \\ &= A V_s - A (\beta V_o) \end{aligned}$$

$$V_o(1 + A\beta) = AV_s$$

So that overall gain with feedback $A_f = \frac{V_o}{V_s} = \frac{A}{1 + A\beta}$

The gain A_f is reduced by the factor $(1 + A\beta)$

Voltage-shunt feedback:

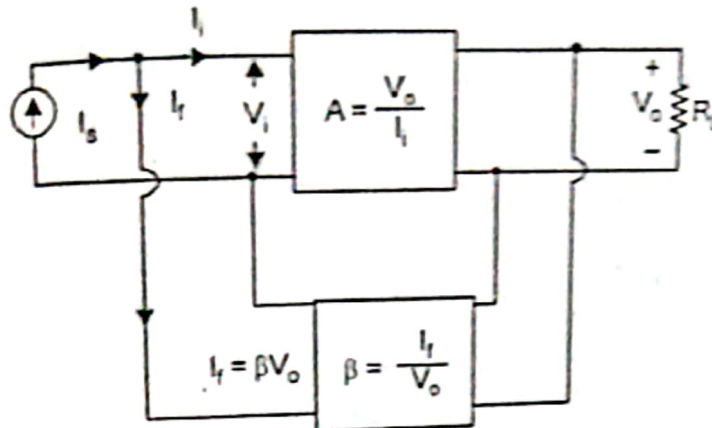


Fig. (b)

$$A_f = \frac{V_o}{I_s} = \frac{AI_i}{I_i + I_f} = \frac{AI_i}{I_i + \beta V_o}$$

$$A_f = \frac{AI_i}{I_i + A\beta I_i} = \frac{AI_i}{(1 + A\beta)I_i}$$

$$A_f = \frac{A}{(1 + A\beta)}$$

Input Impedance with feedback:

⇒ Voltage series feedback: From the connection shown in fig. (a) the input

impedance without feedback $z_i = \frac{V_i}{I_i}$

and input impedance with feedback $z_{if} = \frac{V_s}{I_i}$

$$I_i = \frac{V_i}{Z_i} = \frac{V_s - V_f}{Z_i} = \frac{V_s - \beta V_o}{Z_i}$$

$$I_i = \frac{V_s - A\beta V_i}{Z_i}$$

$$I_i Z_i = V_s - A\beta V_i$$

$$V_s = I_i Z_i + A\beta V_i$$

$$\frac{V_s}{V_i} = Z_i + A\beta \frac{V_i}{I_i} = Z_i + A\beta Z_i = Z_i(1 + A\beta)$$

$$Z_{if} = \frac{V_s}{I_i} = Z_i(1 + A\beta)$$

⇒ Voltage shunt feedback: From the connection shown in Fig. (b)

$$Z_{if} = \frac{V_i}{I_s} = \frac{V_i}{I_i + I_f}$$

$$= \frac{V_i}{I_i + \beta V_o} = \frac{V_i / I_i}{I_i / I_i + \beta V_o / I_i}$$

$$Z_{if} = \frac{Z_i}{(1 + A\beta)} \quad \text{where } A = \frac{V_o}{I_i}$$

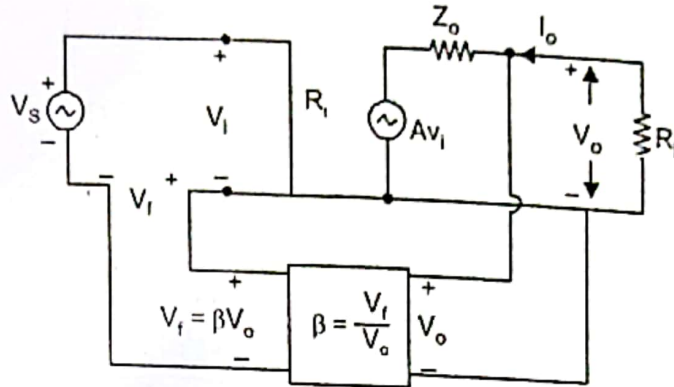
Output impedance with feedback:

Voltage series feedback:

For

$$V_s = 0$$

$$V_i = -V_f$$

Considering V_o as source at o/p

Now

$$V_o = Z_o I_o + A V_i$$

$$V_o = Z_o I_o + A(-V_f)$$

$$V_o = Z_o I_o - A V_f$$

$$V_o = Z_o I_o - A\beta V_o$$

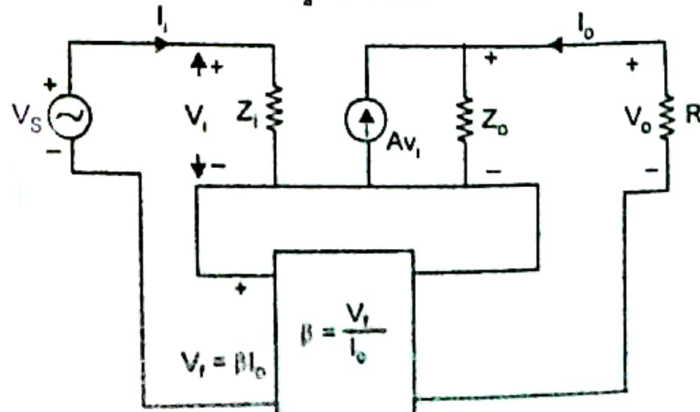
$$V_o(1 + A\beta) = Z_o I_o$$

$$\frac{V_o}{I_o} = \frac{Z_o}{1 + A\beta}$$

$$Z_{of} = \frac{Z_o}{1 + A\beta}$$

where $Z_o \Rightarrow$ output impedance without feedback $Z_{of} \Rightarrow$ output impedance with feedback **$\Rightarrow$ Current Series feedback:**

$$V_s = 0 \text{ and}$$

Considering V_o as source voltage connected at output.

$$V_i = V_f$$

$$I_o = \frac{V_o}{Z_o} - AV_i = \frac{V_o}{Z_o} - AV_f$$

$$I_o = \frac{V_o}{Z_o} - A\beta I_o$$

$$V_o = Z_o I_o (1 + A\beta)$$

$$Z_{of} = \frac{V_o}{I_o} = Z_o (1 + A\beta)$$

or

Effect of feedback connection on input and output impedance

	Voltage Series	Current Series	Voltage Shunt	Current Shunt
Z_{if}	$Z_i(1 + A\beta)$ (increased)	$Z_i(1 + A\beta)$ (increased)	$\frac{Z_i}{(1 + A\beta)}$ (decreased)	$\frac{Z_i}{(1 + A\beta)}$ (decreased)
Z_{of}	$\frac{Z_o}{1 + A\beta}$ (decreased)	$Z_o(1 + A\beta)$ (increased)	$\frac{Z_o}{(1 + A\beta)}$ (decreased)	$Z_o(1 + A\beta)$ (increased)

Q. 5. (b) Enumerate the advantages of negative feedback. (6)

Ans. Negative feedback reduces the gain of the amplifier but improves its performance in several aspects.

Advantages: (i) It improves the gain stability.

The variations in temperature, supply voltages ageing of components or variation in transistor parameters with replacement are some of factors that effect the gain of an amplifier. However, the overall gain of amplifier can be made independent of these variation if negative feedback is used.

Voltage gain with negative feedback

$$A_f = \frac{A}{1 + A\beta} \approx \frac{A}{A\beta} \approx \frac{1}{\beta} \quad (\because A\beta \gg 1)$$

A_f is thus independent of A which depends on passive elements.

Again,

$$A_f = \frac{A}{1 + A\beta}$$

$$\frac{dA_f}{dA} = \frac{(1 + A\beta) - A\beta}{(1 + A\beta)^2} = \frac{1}{(1 + A\beta)^2}$$

$$dA_f = \frac{dA}{(1 + A\beta)^2}$$

and

$$\frac{dA_f}{A_f} = \frac{dA}{A} \times \frac{1 \times A\beta}{(1 + A\beta)^2}$$

$$\frac{dA_f}{A_f} = \frac{1}{1 + A\beta} \cdot \frac{dA}{A}$$

in negative feedback $(1 + A\beta) \gg 1$, the percentage change in gain with negative feedback is less than the percentage change in gain without feedback. Thus negative feedback improves the gain stability of the amplifier.

(ii) **It reduces non-linear distortion and noise:** The distortion takes place due to non-linearity in the transfer characteristics of transistor or FET. Due to the introduction of negative feedback reduces the non-linear distortion. The total distortion in negative feedback amplifier can be given as

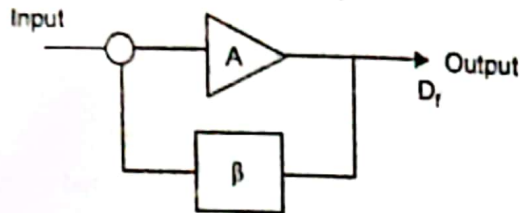
$$D_f = D - A\beta D_t$$

where D is the distortion without feedback

$$D_f(1 + A\beta) = D$$

$$D_f = \frac{D}{(1 + A\beta)}$$

thus the distortion reduces by the factor $(1 + A\beta)$



(iii) **It increases the bandwidth:**

The gain with negative feedback $A_f = \frac{A_{VO}}{1 + A_{VO}\beta}$

Where A_{VO} is gain without feedback.

The gain decreases with introduction of negative feedback. The amplifiers exhibit a constant gain-bandwidth product. The reduction in gain is therefore, accompanied by increase in bandwidth. The bandwidth increases by the same factor $(1 + A_{VO}\beta)$ by which the gain reduces as shown in the figure.

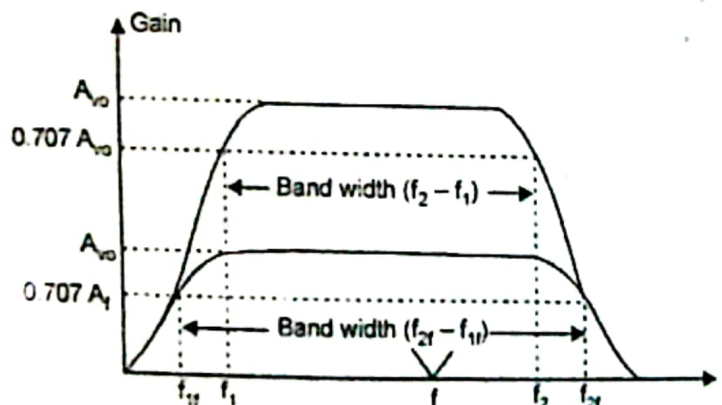


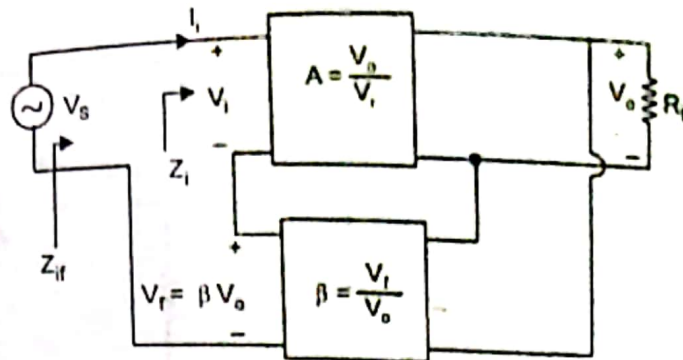
Fig. Voltage gain is decreased and bandwidth is increased for an amplifier using negative feedback

(iv) **It increases input impedance (Z_{if})**

$$I_i = \frac{V_i}{Z_i} = \frac{V_s - V_f}{Z_i}$$

$$= \frac{V_s - \beta V_o}{Z_i}$$

$$I_i = \frac{V_s - A\beta V_i}{Z_i}$$



$$I_i Z_i = V_s - A\beta V_i$$

$$V_s = I_i Z_i + A\beta V_i$$

$$\frac{V_s}{I_i} = Z_i + A\beta \frac{V_i}{I_i}$$

or

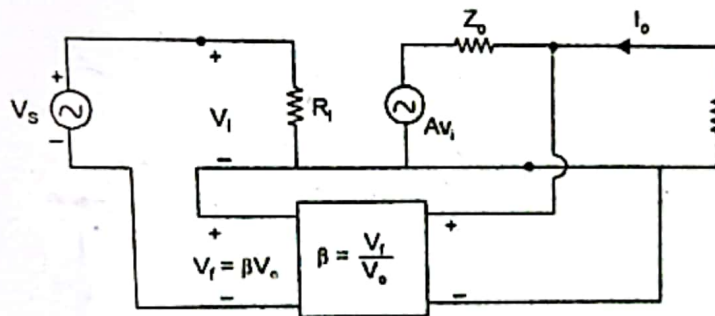
$$Z_{if} = Z_i + A\beta Z_i$$

$$Z_{if} = Z_i (1 + A\beta)$$

Thus overall input impedance Z_{if} is increased by the factor $(1 + A\beta)$

(v) It reduces output impedance (Z_{of})

For $V_s = 0$ and considering V_o as a source voltage connected at output.



Now

$$V_o = Z_o I_o + A V_i$$

$$V_o = Z_o I_o + A(-V_f)$$

$$V_o = Z_o I_o - A V_f$$

$$V_o = Z_o I_o - A\beta V_o$$

$$V_o (1 + A\beta) = Z_o I_o$$

$$\frac{V_o}{I_o} = \frac{Z_o}{(1 + A\beta)}$$

or

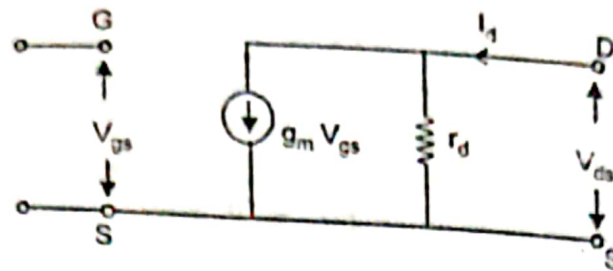
$$Z_{of} = \frac{V_o}{I_o} = \frac{Z_o}{1 + A\beta}$$

$$Z_{of} = \frac{Z_o}{(1 + A\beta)}$$

Q. 6. (a) Draw the small signal model of FET and derive an expression for A_v and R_o in CD configuration. (8.5)

Ans. In low frequency small signal model, as the gate is reverse biased, the gate current becomes zero. So, the gate to source junction is represented by an open circuit and no current is drawn by the input terminal of the FET. The reason is that input

resistance is very large. The output side behaves as a current source whose magnitude is controlled by V_{gs} and g_m



The drain current

$$I_d = g_m V_{gs} + \frac{1}{r_d} V_{ds}$$

Common Drain JFET Amplifier, OR source follower:

In the common drain circuit, the output voltage is developed across source resistor R_s . It is seen that the output voltage from a common drain circuit is approximately the same as the input voltage. Thus this circuit has approximately unity gain.

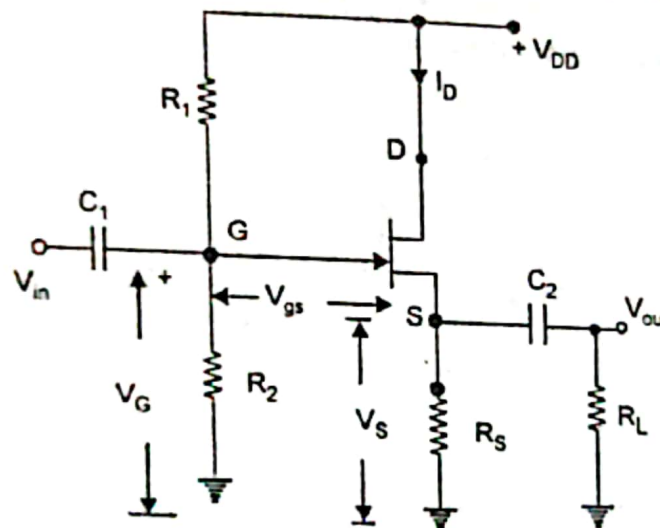


Fig. (a) Common drain amplifier circuit.

That is why, this is also known as source follower. The common drain ac equivalent circuit is shown in fig. (b). The current generator is $g_m V_{gs}$, where $V_{gs} = V_{in} - V_{out}$ and

$$R_G = R_1 \parallel R_2$$

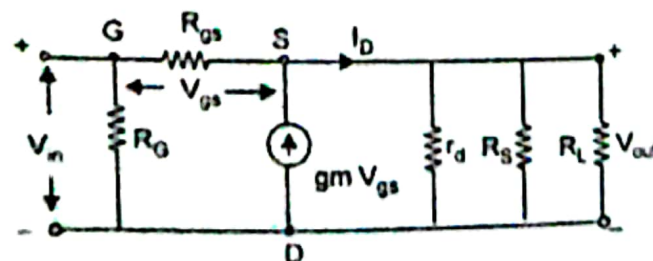


Fig. (b)

$$V_{out} = I_D (r_d \parallel R_s \parallel R_L)$$

$$V_{out} = g_m V_{gs} (r_d \parallel R_s \parallel R_L)$$

$$V_{in} = V_{gs} + V_{out}$$

$$= V_{gs} + g_m V_{gs} (r_d \parallel R_s \parallel R_L)$$

or
and

Voltage gain

$$A_v = \frac{V_{out}}{V_{in}} = \frac{g_m V_{gs} (r_d \parallel R_S \parallel R_L)}{V_{gs} (1 + g_m (r_d \parallel R_S \parallel R_L))}$$

$$A_v = \frac{g_m (r_d \parallel R_S \parallel R_L)}{1 + g_m (r_d \parallel R_S \parallel R_L)}$$

normally

$$r_d \gg R_S \parallel R_L$$

∴

$$A_v = \frac{g_m (R_S \parallel R_L)}{1 + g_m (R_S \parallel R_L)}$$

if

$$g_m (R_S \parallel R_L) \gg 1$$

then

$$A_v \approx 1$$

⇒ Input impedance is $R_G = R_1 \parallel R_2$ ⇒ **Output impedance:** The voltage V_{out} affects upon the gate - source voltage V_{gs} and V_{gs} is given as

$$V_{gs} = V_{out} \times \frac{R_{gs}}{(R_S \parallel R_G) + R_{gs}}$$

Normally

$$R_{gs} \gg R_S \parallel R_G$$

∴

$$V_{gs} = V_{out}$$

and

$$I_D = g_m V_{gs}$$

$$Z_S = \frac{V_{out}}{I_D} = \frac{V_{gs}}{g_m V_{gs}} = \frac{1}{g_m}$$

Actually, r_d is in parallel with $\frac{1}{g_m}$, but $r_d \gg \frac{1}{g_m}$ it can be neglected. R_s in the output impedance of the device

$$Z_{out} = R_s \parallel \frac{1}{g_m}$$

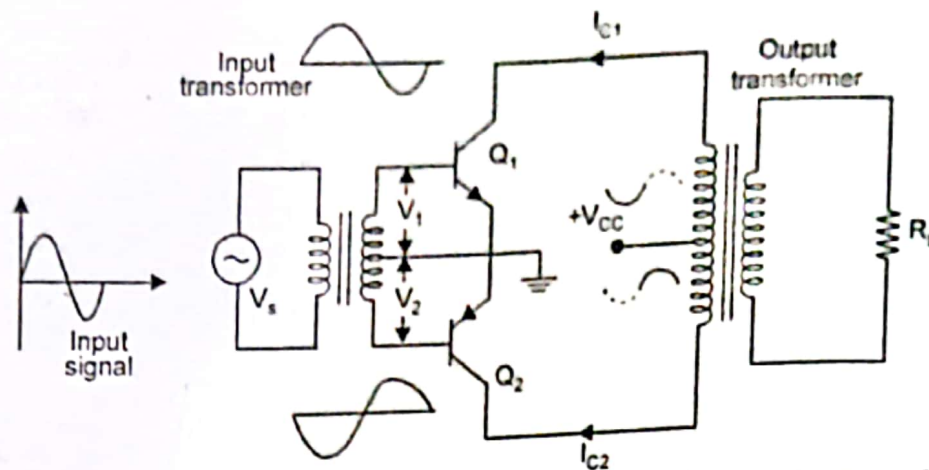
Q. 6. (b) What is pinch off voltage? Explain its significance. (4)

Ans. Pinch off voltage: The value of drain-source voltage (V_{ds}) at which channel is pinched-off (All the free charges from the channel get removed) is called pinch-off voltage.

The amount of reverse bias is not the same throughout the length the p-n junction. Thus the channel becomes narrower at the drain end in comparison to that at source end. The channel is not completely closed at the drain end. Because in that case there will be no drain current, so there will be no voltage drop along the channel length and amount of reverse bias will become uniform and wedge-shaped depletion region will become rectangular one. The channel will open and the drain current will flow. However, at pinch-off voltage, the channel width is reduce to a constant minimum value allow the flow of drain current.

Q. 7. (a) Explain how push pull configuration is used in power amplifiers? Why and where it is used. (6.5)

Ans. Class B push-pull operation is the same as that for the class A operation except that the devices are biased at cut-off. It gives higher operating efficiency (theoretical 78.5% due to the fact that no power is drawn by the circuit under zero-signal condition.



Circuit operation: When the input signal is applied the centre tapped secondary of the input transformer develops two signals which are identical but in phase opposition. The transistor Q_1 and Q_2 are driven by these two signals. Thus when V_1 is going positive, V_2 is going negative, so that transistor Q_2 is being biased further off when transistor Q_1 is being biased ON. As the collector current in Q_1 increases from zero, it produces a half sine wave of voltage across upper half of the primary of the output transformer. When the positive half cycle of the input signal to Q_1 begins to go negative, the signal at Q_2 base is commencing to go positive. Thus Q_1 becomes biased off again, Q_2 is biased ON and a half cycle of voltage waveform is generated across the lower primary winding of output transformer.

The effect of the two-half cycles in separate halves of the primary of the output transformer is to produce a magnetic flux in the transformer core, which flows first in one direction and then in the opposite direction. This flux links with the secondary of the output transformer and generates a complete sine-wave output, which is passed on to the load. Thus, in the class B circuit, the two transistors are said to operating in push-pull.

Q. 7. (b) Write a short note on tuned amplifiers: Also explain how its efficiency will be calculated. (6)

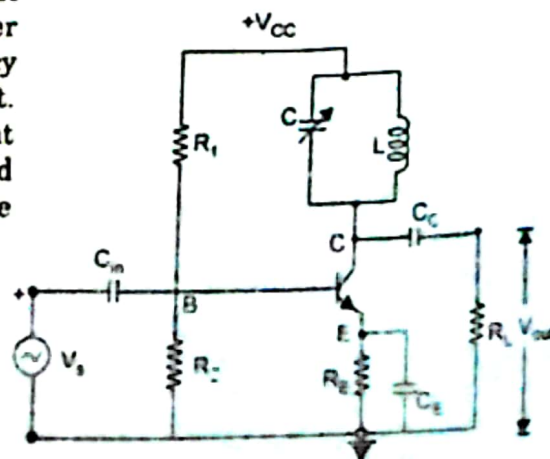
Ans. Tuned amplifier selects and amplifies a single frequency while audio frequencies are mixtures of frequencies. Small signal tuned amplifier may be classified into single tuned and double tuned amplifiers. In a single tuned amplifier, one parallel tuned circuit is employed while in double tuned amplifier, two inductively coupled tuned circuit are employed per stage and both the tuned circuits being tuned to the same frequency.

Single Tuned Amplifiers: The circuit arrangement is shown in the figure. A parallel tuned L-C circuit connected in the collector circuit, the impedance of which depends upon the frequency, acts as a collector load. Capacitor C generally variable so that the resonant frequency of the circuit may be varied. The tuned circuit will offer very high impedance to the signal frequency and thus large output will appear across it. The frequency corresponding to the resonant frequency of tuned circuit will be amplified and other frequencies will be rejected by the tuned circuit.

The resonant frequency

$$f_r = \frac{1}{2\pi\sqrt{LC}}$$

Gain $A_V = \frac{\beta R_{ac}}{r_{in}}$



where R_{ac} tuned circuit impedance

$$R_{ac} = Z = \frac{L}{CR}$$

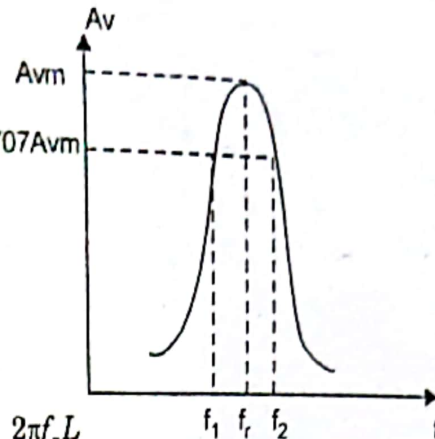
and r_m is input resistance
 $\Rightarrow$ Band width

$$BW = f_2 - f_1$$

$$B.W. = \frac{f_r}{Q}$$

$\Rightarrow$ Quality factor (Q)

$$Q = \frac{X_L}{R} = \frac{WL}{R} = \frac{2\pi f_r L}{R}$$



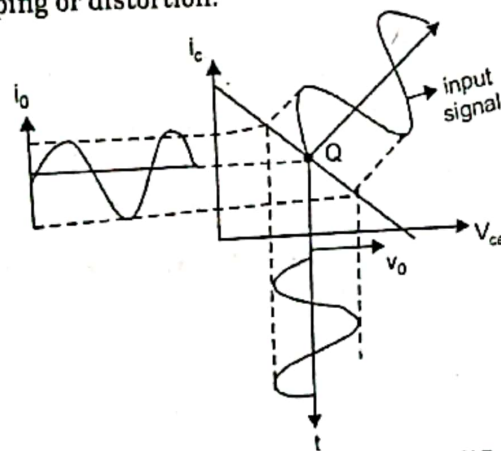
Higher quality factor Q, the more selective is the tuned circuit. A high Q-circuit will give a high voltage gain but at the same time it will give much reduced band width because the bandwidth is inversely proportional to the Q factor.

Q. 8. (a) Explain the construction and characteristics of SCR. (6)

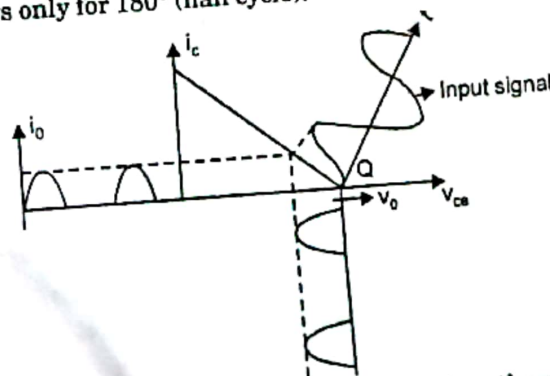
Ans. Refer Q. 7. (a) End Term Examination 2017.

Q. 8. (b) Compare class A, B and C amplifier. (6.5)

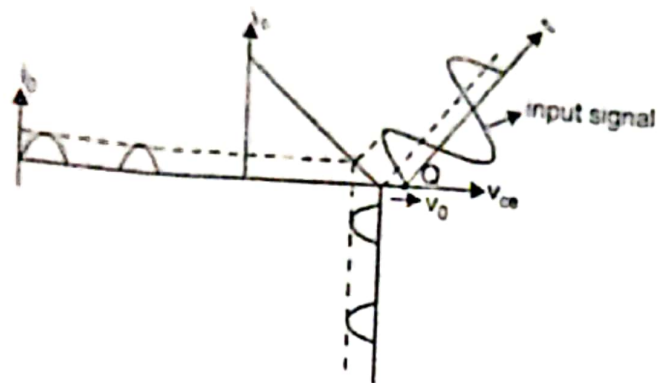
Ans. (i) Class A Power Amplifier: When an amplifier is so biased that Q point lies at the centre of load line in active region. Output current flows for all the 360° of the cycle without any clipping or distortion.



(ii) Class B Power Amplifier: When Q point of amplifier is fixed at cut off and output current flows only for 180° (half cycle).



(iii) Class C Power Amplifier: Q point is fixed below the cut off point then some positive part is lost and output current flows only for less than 180° .



Q. 9. Write short

(a) Tuned Amplifier

Ans. Refer Q. 7. (b) End Term Examination 2018.

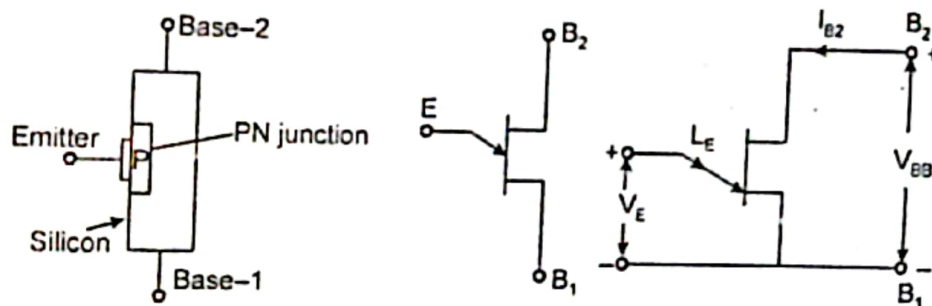
(6)

Q. 9. (b) UJT

(6.5)

Ans. UJT (Unijunction Transistor): It is also called the double base diode is a 2 layer 3 terminal solid state (silicon) switching device. The device has a unique characteristic that when it is triggered, its I_E increases regeneratively until it is restricted by emitter power supply.

V-I characteristics:



Basic structure, Symbol, Basic arrangement

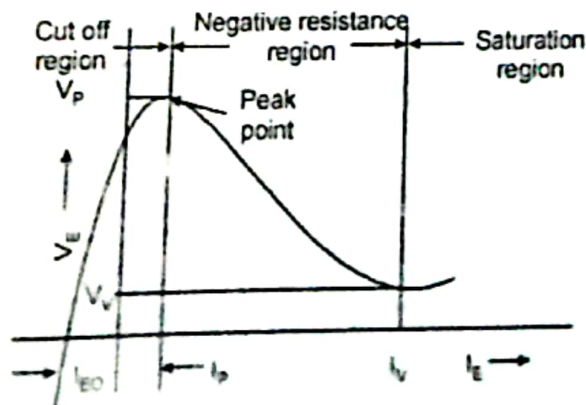
The device has only one Junction, so it is called the unijunction device. Its structure is similar to N-channel JFET.

UJT is operated with Emitter $J^n FB$ while the JFET is normally operated with gate $J^n RB$.

It exhibits a negative resistance characteristic and so it can be employed as an oscillator. It has high pulse current capability.

A very low value of triggering current lost cost.

A stable triggering voltage (V_P)-a fixed fraction of applied inter base voltage V_{BB} .



Emitter Characteristic of a UJT

END TERM EXAMINATION [DEC. 2017] THIRD SEMESTER [B.TECH] ANALOG ELECTRONICS-I [ETEC-203]

M.M.: 75

Time : 3 hrs.

Note: Attempt any five questions including Q. no. 1 which is compulsory. Select one question from each unit. Assume missing data, if any.

Q.1. Attempt all the questions:

Q.1. (a) If two transistors are connected back to back will it work as a transistor? (2.5)

Ans. Two diodes connected back to back will not work as transistor because of base width much larger than the diffusion length of minority carriers in the base. All minority carriers will recombine in the base and none will reach the collector. For this case the current gain α will be zero. Under these conditions, transistor action ceases, and we simply have two diodes placed back to back. This type of connection is known as Ebers-Moll model.

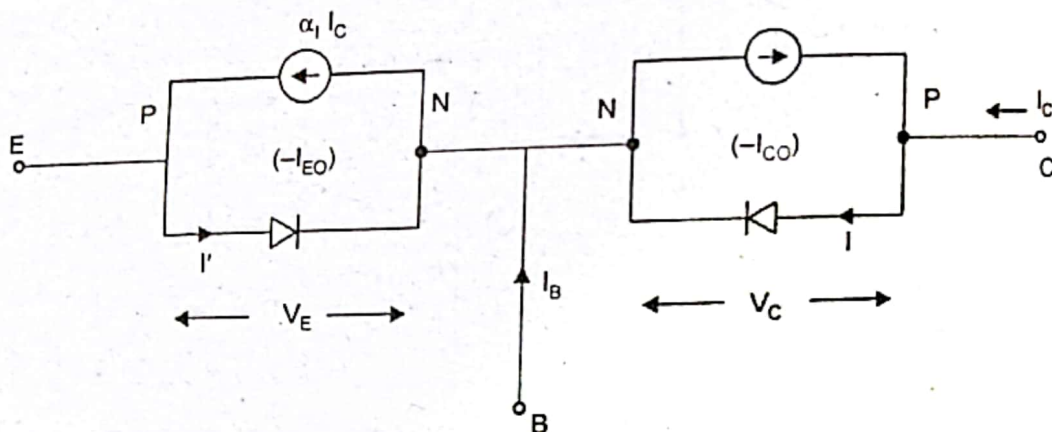


Fig. Ebers Moll Model for P-N-P Transistor

Q.1. (b) What is the significance of stability factor? Define $S(I_{CO})$, $S_{(\beta_{dc})}$ and $S(V_{BE})$. (2.5)

Ans. **Stability Factor:** The stability of Q point of a transistor amplifier depends on the following three parameters:

1. Leakage current I_{CO}
2. β_{dc}
3. Base to emitter voltage V_{BE}

The effect of these parameters can be expressed mathematically by defining the stability factors for the three parameters individually as follows:

$$(i) \text{ Stability factor, } S = \frac{\Delta I_C}{\Delta I_{CO}} \text{ Constant } V_{BE} \text{ and } \beta_{dc} \text{ or } \frac{\partial I_C}{\partial I_{CO}} \quad \dots (1)$$

This represents the change in collector current due to change in reverse saturation current I_{CO} . The other two parameters that means V_{BE} and β_{dc} are assumed to be constant.

(ii) Stability factor, $S' = \frac{\Delta I_C}{\Delta V_{BE}}$ Constant I_{CO} and β_{dc} or $\frac{\Delta I_C}{\Delta V_{BE}}$ (2)

S' represents the change in I_C due to change in V_{BE} at constant I_{CO} and β_{dc} .

(iii) Stability factor, $S'' = \frac{\Delta I_C}{\Delta \beta_{dc}}$ Constant I_{CO} and V_{BE} or $\frac{\Delta I_C}{\Delta \beta_{dc}}$ (3)

These expression indicate that I_C is collectively dependent on ΔI_{CO} , $\Delta \beta_{dc}$. The combined effect to these parameters on I_C is mathematically given by.

Total change in collector current
 $\Delta I_C = S \Delta I_{CO} + S' \Delta V_{BE} + S'' \Delta \beta_{dc}$

(4)

Ideally the values of all the stability factors should be zero and practically they should be as small as possible.

Practically the value of S is significantly higher than the other two stability factors. Hence while comparing the biasing circuits, we should focus our attention more on the value of S .

Q.1. (c) Explain the function of coupling capacitor and bypass capacitor. (2.5)

Ans. The coupling capacitor C_c transmits ac signal but blocks the dc voltage of the first stage from reaching the base of the second stage. Thus the dc biasing of the next stage is not interfered.

The Emitter by pass capacitor C_E , offers low reactance path to the signal. If it is not present, then the voltage drop across R_E will reduce the effective voltage available across the base-emitter terminals and thus reduce the gain.

Q.1. (d) Why the channel of a JFET is never completely closed at the drain end? (2.5)

Ans. The Channel is not completely closed at the drain end, because in that case there will be no drain current, so there will be no voltage drop along the channel length and amount of reverse bias will become uniform and the wedge-shaped depletion region will become rectangular one.

Q.1. (e) Define "current controlled device" and "voltage controlled device"? (2.5)

Ans. A current controlled device is one in which current defines the operating conditions of the device, whereas a voltage controlled device is one in which a particular voltage defines the operating conditions.

Q.1. (f) Why FET is called a voltage controlled device? (2.5)

Ans. In a FET, drain current is controlled by the effect of the extension of the field associated with the depletion region developed by the reverse bias on the gate, so it is called a voltage controlled device.

Q.1. (g) What is negative resistance region in UJT? Explain its physical significance. (2.5)

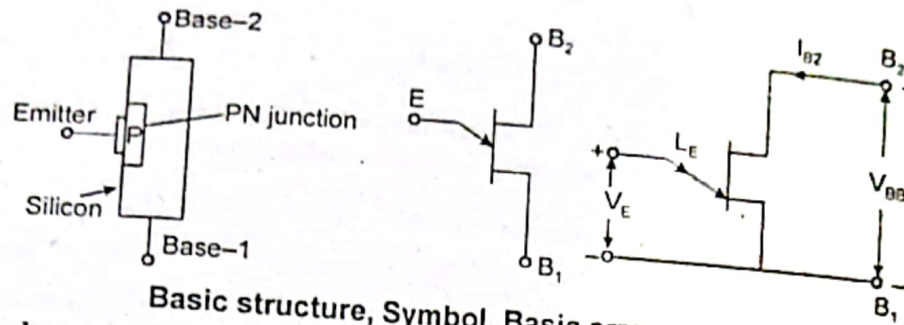
Ans. UJT (Unijunction Transistor): It is also called the double base diode is a 2 layer 3 terminal solid state (silicon) switching device. The device has a unique characteristic that when it is triggered, its I_E increases regeneratively until it is restricted by emitter power supply.

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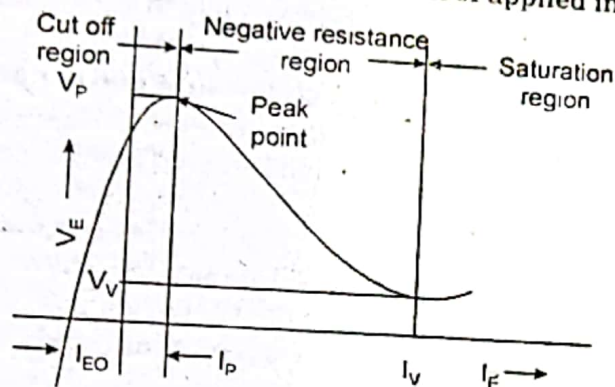
V-I characteristics:



Basic structure, Symbol, Basic arrangement

A very low value of triggering current lost cost.

A stable triggering voltage (V_P)-a fixed fraction of applied inter base voltage V_{BB}.



Emitter Characteristic of a UJT

Q.1. (h) How does negative feedback increases bandwidth of an amplifier?

Ans. The gain with negative feedback $V_{vf} = \frac{A_{V0}}{1 + A_{V0}}$

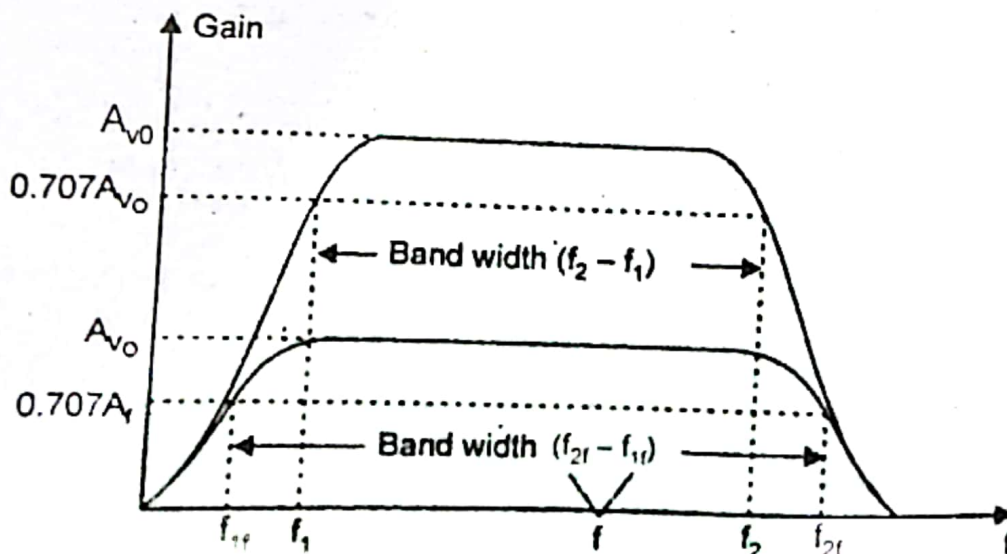


Fig. Voltage gain is decreased and bandwidth is increased for an amplifier using negative feedback.

Where A_{v_0} is gain without feedback.

The gain decreases with introduction of negative feedback. The amplifiers exhibit a constant gain-bandwidth product. The reduction in gain is therefore, accompanied by increase in bandwidth. The bandwidth increases by the same factor $(1 + A \beta)$ by which the gain reduces as shown in the figure.

Q.1. (i) What advantages does the transformer provide in a transformer coupled power amplifier?

Ans. Transformer provides the following advantages in a transformer coupled power amplifier. (2.5)

1. Power loss in the primary of the transformer used for coupling the load to the amplifier, is negligible and therefore conversion efficiency is doubled in a transformer coupled power amplifier.

2. Output transformer makes the transfer of maximum power to the load through impedance matching.

3. Output transformer prevents a large dc current from flowing in the load. Which otherwise could be harmful if the load were a loudspeaker.

Q.1. (j) What are the drawbacks of complementary symmetry push pull class B power amplifier?

Ans. One obvious drawback of this circuit arrangement is the requirement for two supply voltages. Another drawback is difficult of obtaining matched complementary transistors. If there is an unbalance in the characteristics of the two transistors, even harmonic will no longer be cancelled and considerable distortion will be introduced.

UNIT - I

Q.2. (a) A Si transistor has a potential divider biasing. If it operates at $I_C = 3\text{mA}$ and $V_{CE} = 10\text{V}$. Find V_{CC} , R_B , V_B and R_E for $\beta = 100$. Assume $V_{BE} = 0.7\text{V}$. (6)

Ans. $I_C = \beta I_B$

$$I_B = \frac{I_C}{\beta} = \frac{3 \times 10^{-3}}{100} = 0.03\text{mA}$$

$$I_1 = 10 I_B = 10 \times 0.03 = 0.3\text{mA}$$

The value of R_{B1} is so decided that the current I flowing through R_{B1} and R_{B2} ($20\text{K}\Omega$) is about 10 times of the base current to achieve good stabilization by the circuit shown

$$V_{CB} + V_{BE} = V_{CE}$$

$$V_{CB} = V_{CE} - V_{BE} = 10 - 0.7$$

$$= 9.3\text{ Volt}$$

and $I_1 R_{B1} = I_C R_C + V_{CB}$

$$= 3 \times 10^{-3} \times 5 \times 10^3 + 9.3$$

$$= 24.3\text{ V}$$

$$R_{B1} = \frac{24.3}{I_1} = \frac{24.3}{0.3 \times 10^{-3}}$$

$$R_{B1} = 81\text{ K}\Omega$$

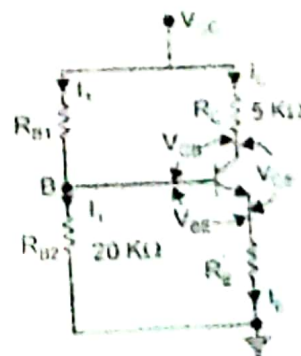
$$V_B = I_1 \times R_{B2} = 0.3 \times 10^{-3} \times 20 \times 10^3$$

$$V_B = 6\text{ Volt}$$

$$V_{CC} = I_1 (R_{B1} + R_{B2})$$

$$= 0.3 \times 10^{-3} (81 + 20) \times 10^3$$

$$V_{CC} = 30.3\text{ Volt}$$



Now

$$I_E R_E = V_B - V_{BE}$$

$$I_E R_E = 6 - 0.7 = 5.3 \text{ Volt}$$

$$R_E = \frac{5.3}{I_E} = \frac{5.3}{3 \times 10^{-3}}$$

$$R_E = 1.77 \text{ K}\Omega \text{ (Assuming } I_C \approx I_E)$$

Q.2. (b) Explain how phase reversal of the signal takes place when it is amplified by a single stage CE voltage amplifier? (2.5)

Ans. In a CE configuration, the output voltage increases in the negative direction when the input signal voltage increases in the positive direction and vice-versa. This is called the phase reversal and causes a phase difference of 180° between the input signal voltage and output voltage

Q.2. (c) How many types of biasing are done on a BJT to work properly as an amplifier? Which one is the best and why? (4)

Ans. Following are the types of biasing technique used on a BJT.

- (i) Fixed bias
- (ii) Collector to base bias
- (iii) Fixed bias with emitter resistor
- (iv) Voltage divider or self bias circuit.

Voltage divider or self bias circuit is the most commonly used biasing arrangement. Its popularity is primarily due to its low sensitivity to variations in β from one transistor to another of the same lot.

Q.3. (a) Derive expression for voltage gain, current gain, input resistance and output resistance of a common base Si transistor circuit using h parameters. (7.5)

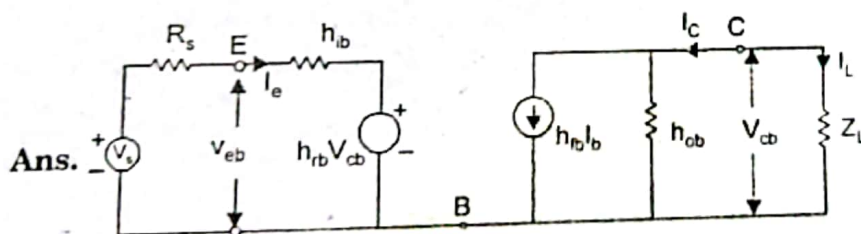


Fig. Hybrid model of common base transistor amplifier

Current gain

$$A_I = \frac{I_L}{I_e} = -\frac{I_c}{I_e}$$

But

$$I_c = h_{fb} I_b + h_{ob} V_{cb}$$

or

$$I_c = h_{fb} I_b + h_{ob} (-I_e Z_L)$$

$$I_c = h_{fb} I_b - h_{ob} h_{fb} I_b Z_L$$

$$I_c = h_{fb} I_b (1 - h_{ob} Z_L)$$

and

$$I_e = I_c + I_b = h_{fb} I_b + I_b$$

$$I_e = I_b (h_{fb} + 1)$$

$$A_I = -\frac{I_c}{I_e} = \frac{h_{fb} I_b (1 - h_{ob} Z_L)}{I_b (h_{fb} + 1)}$$

$$A_I = \frac{h_{fb} (1 - h_{ob} Z_L)}{(h_{fb} + 1)}$$

The input impedance (Z_i)

($\because h_{fb} \gg 1$)

$$Z_i = \frac{V_{eb}}{I_b}$$

$$\begin{aligned} V_{eb} &= h_{ib} I_e + h_{rb} V_{cb} \\ &= h_{ib} (h_{fb} + 1) I_b + h_{rb} h_{fb} I_b Z_L \end{aligned}$$

$$\frac{V_{eb}}{I_b} = h_{ib} (h_{fb} + 1) + h_{rb} h_{fb} Z_L \quad (\because h_{fb} \gg 1)$$

$$Z_i = h_{ib} (h_{fb} + 1)$$

Voltage Gain (A_v)

$$A_v = \frac{V_{cb}}{V_{eb}} = \frac{h_{fb} I_b Z_L}{h_{ib} h_{fb}}$$

$$A_v = \frac{I_b Z_L}{h_{ib}}$$

Output Impedance (Z_o)

$$Z_o = \frac{V_{cb}}{I_c} = \frac{-I_c Z_L}{I_c}$$

$$Z_o = Z_L$$

Q.3. (b) What is Darlington pair? Derive an expression for current gain in Darlington pair. (5)

Ans. Refer to Q.1 (c), First Term Examination, 2016.

UNIT - II

Q.4. (a) Draw the cascade of a common emitter stage and common collector stage and give the expression for input resistance and current gain of the circuit. (7.5)

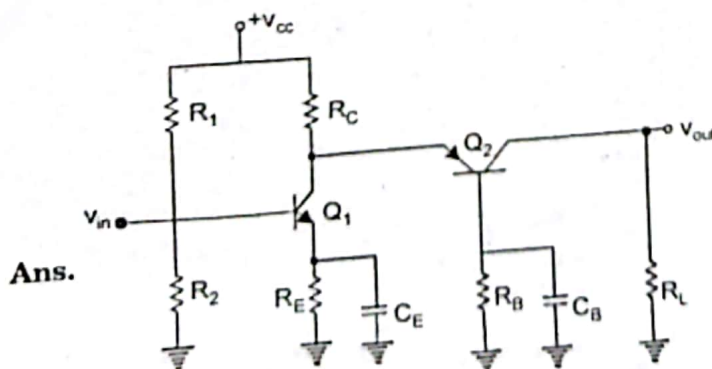


Fig. Cascade Amplifier

This arrangement is designed to provide a high input impedance with low voltage gain to ensure that input miller capacitance is at a minimum with CB stage providing good high frequency operation.

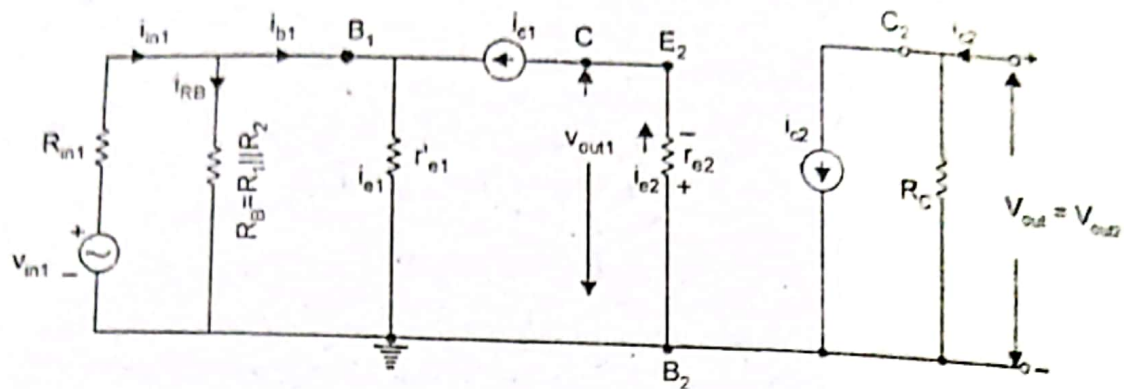


Fig. Small signal T-equivalent circuit of cascade amplifier

Current Gain:

The current gain A_i of cascade (CE-CB) configuration may be expressed as

$$\begin{aligned}
 A_i &= \frac{i_{out}}{i_{in1}} = \frac{i_{b1}}{i_{in1}} \times \frac{i_{e1}}{i_{b1}} \times \frac{i_{e2}}{i_{e1}} \times \frac{i_{c2}}{i_{e2}} \times \frac{i_{out}}{i_{c2}} \\
 &= \frac{i_{b1}}{i_{in1}} \times \frac{i_{e1}}{i_{b1}} \quad (\because i_{e1} = i_{c1} = i_{e2} = i_{c2} = i_{out})
 \end{aligned}$$

Applying current divider rule to the input circuit

$$i_{b1} = \frac{R_B i_{in1}}{R_B + \beta_{ac} r_{e1}} \quad (\because R_B = R_1 \parallel R_2)$$

Also

$$i_{e1} = \beta_{ac} i_{b1}$$

$$\therefore A_i = \frac{\beta_{ac} R_B}{R_B + \beta_{ac} r_{e1}}$$

Input resistance:

$$V_{in1} = r_{e1} i_{e1} = \beta_{ac} r_{e1} i_{b1} \quad (\because i_{e1} = \beta_{ac} i_{b1})$$

$$V_{in1} = \beta_{ac} r_{e1} \times \frac{R_B i_{in1}}{R_B + \beta_{ac} r_{e1}}$$

$$\text{and input resistance } R_{in} = \frac{V_{in1}}{i_{in1}}$$

$$R_{in} = \frac{\beta_{ac} r_{e1} R_B}{R_B + \beta_{ac} r_{e1}}$$

Q.4. (b) A single stage amplifier has a gain of 60. The collector load $R_C = 500 \Omega$ and the input impedance is $1 \text{ K}\Omega$, calculate the overall voltage gain when two stages are cascaded through RC coupling.

Ans. In case of mid-frequency range the overall voltage gain

$$A_{v_{total}} = \frac{h_{fe} R_C}{R_i + h_{ie}}$$

$$A_{em} = \frac{60 \times 500}{500 + 1000} = \frac{30000}{1500} = 20.$$

Q.5. (a) How the negative feedback changes the impedances? Explain with mathematical expression. (7.5)

Ans. Refer to Q.5 (a), End Term Examination, 2016.

Q.5. (b) Why voltage series feedback is most commonly used in cascaded amplifier? (2)

Ans. In cascaded amplifiers, high input impedance and low output impedance are usually the main requirements and both these requirements can be met by voltage series feedback, so it is most commonly used in cascaded amplifiers.

Q.5. (c) The open loop gain of an amplifier changes by 5%. If 10 dB negative feedback is applied, calculate the percentage change in the closed loop gain. (3)

Ans. Percentage change in closed loop gain

$$\frac{dA_f}{A_f} = \frac{dA}{A} \frac{1}{1 + A_p} \times 100$$

Where

$$\frac{dA}{A} = 0.05 \text{ and } A_p = 10\text{dB} = 3.16$$

$$\frac{dA_f}{A_f} = \frac{0.05}{1 + 3.16} \times 100 = \frac{0.05}{4.16} \times 100 = 1.2\%.$$

UNIT-III

Q.6. (a) Explain the meaning of pinch off voltage in a JFET. How does the current flow in a JFET after pinch off? Also explain the meaning of the terminologies: (6)

(i) Trans-conductance

(ii) Drain Resistance and

(iii) Amplification Factor.

Ans. Pinch off voltage: The value of drain-source voltage (V_{ds}) at which channel is pinched-off (All the free charges from the channel get removed) is called pinch-off voltage.

The amount of reverse bias is not the same throughout the length the p-n junction. Thus the channel becomes narrower at the drain end in comparison to that at source end. The channel is not completely closed at the drain end. Because in that case there will be no drain current, so there will be no voltage drop along the channel length and amount of reverse bias will become uniform and wedge-shaped depletion region will become rectangular one. The channel will open and the drain current will flow. However, at pinch-off voltage, the channel width is reduced to a constant minimum value to allow the flow of drain current.

Dynamic Drain Resistance (r_d): Drain resistance is an AC resistance of JFET It is defined as the ratio of change in the drain to source voltage to the corresponding change in the drain current, at a constant value of gate to source voltage. Drain resistance is calculated in the saturation region of the FET output characteristics.

$$\left. \frac{\Delta V_{DS}}{\Delta I_D} \right|_{V_{GS} \text{ is constant}}$$

Trans conductance (g_m): Trans conductance g_m is defined as the ratio of change in drain current to the corresponding change in gate to source voltage, at a constant value of drain to source voltage.

$$g_m = \left. \frac{\Delta I_D}{\Delta V_{GS}} \right|_{V_{DS} \text{ is constant}}$$

Amplification factor (μ): Amplification factor ' μ ' is defined as the ratio of change in the drain to source voltage, to change in the gate to source voltage, at a constant value of drain current I_D .

$$\mu = \left. \frac{\Delta V_{DS}}{\Delta V_{GS}} \right|_{I_D \text{ is constant}}$$

Relation between r_d , g_m and μ :

$$\mu = \frac{\Delta V_{DS}}{\Delta V_{GS}}$$

or

$$\mu = \frac{\Delta V_{DS}}{\Delta I_D} \times \frac{\Delta I_D}{\Delta V_{GS}}$$

$$\boxed{\mu = r_d \times g_m}$$

Q.6. (b) Calculate the minimum value of V_{DS} required for an NMOSFET to operate in pinch off condition when $V_{GS} = 1V$, $V_p = -2V$, $I_{DSS} = 10 \text{ mA}$. Also find the corresponding drain current. (4)

Ans.

$$I_D = I_{DSS} \left[1 - \frac{V_{GS}}{V_p} \right]^2$$

$$= 10 \times 10^{-3} \left[1 + \frac{1}{2} \right]^2$$

$$= 10 \times 10^{-3} \left[\frac{3}{2} \right]^2 = 10 \times 10^{-3} \times \frac{9}{4}$$

$$I_D = 22.5 \times 10^{-3} \text{ A}$$

$$I_D = 22.5 \text{ mA}$$

Q.6. (c) Give a comparison between JFET and MOSFET. (2.5)

Ans. JFET and MOSFETs are quite similar in their operating principles and their electrical characteristics. However, they differ in some aspects, as detailed below:

1. In MOSFETs (D-MOSFETs as well as E-MOSFETs), the transverse electric field induced across an insulating layer deposited on the semiconductor material controls the conductivity of the channel where as in the JFETs the transverse electric field across the reverse biased PN junction controls the conductivity of the channel

2. JFETs can only be operated in depletion mode whereas MOSFETs can be operated in their depletion or in enhancement mode

1. MOSFETs have input impedance much higher than that of JFETs because of small leakage current.
2. JFETs have characteristics curves more flatter than those of MOSFETs indicating higher drain resistance.
3. MOSFET is very susceptible to overload voltage and needs special handling during installation.
4. Capacitive effects may be considerably lower in dual gate MOSFETs.
5. Special digital CMOS circuits are available which involve near zero power dissipation and very low voltage and current requirements.
6. When JFET is operated with a reverse bias on the junction the gate current I_G is larger than it would be in a comparable MOSFET.
7. MOSFETs are some what easier to manufacture, they are more widely used than the JFETs.

Q.7. (a) Sketch and explain the silicon controlled Rectifier characteristics with the switching action. (7.5)

Ans. SCR: Silicon Controlled Rectifier is constructed of a semiconductor material with a third terminal for control purpose.

Silicon was chosen because of its high temperature and power capabilities. It is a 3 terminal device, Anode, cathode and gate. Gate determines when the rectifier swides from the open circuit to short circuit gate. It is not enough simply to forward bias the anode to cathode region of the device. The dynamic resistance of the SCR is typically 0.01 to 0.1 Ω and reverse resistance is typically 100 K Ω or were.

It is widely used as a switching devices is power control applications. It can control loads by switching ON and OFF upto many thousand times a second. It can switch ON for a variable length of time duration, thereby delivering desired amount of power to the load. Thus, it possesses the advantage of a cheostat as well as a switch with have of their drawback.

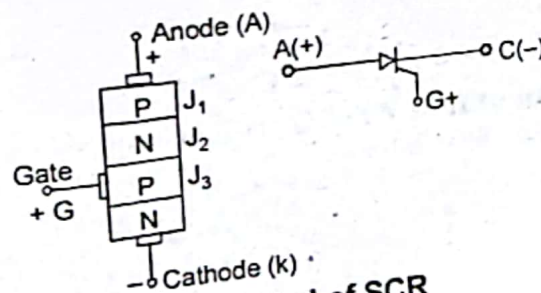


Fig. Symbol of SCR

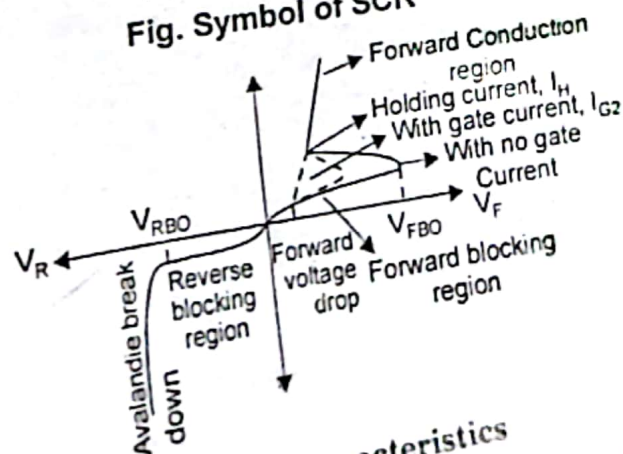


Fig. Characteristics

Q.7. (b) A UJT relaxation oscillator has $R_E = 60 \text{ K}\Omega$ and $C = 0.25 \mu\text{F}$. Determine the pulse repetition frequency. Take intrinsic stand-off ratio to be 0.65? (5)

Ans. Pulse repetition frequency $f = \frac{1}{2.3 R_E C \log_{10} \frac{1}{1-n}}$

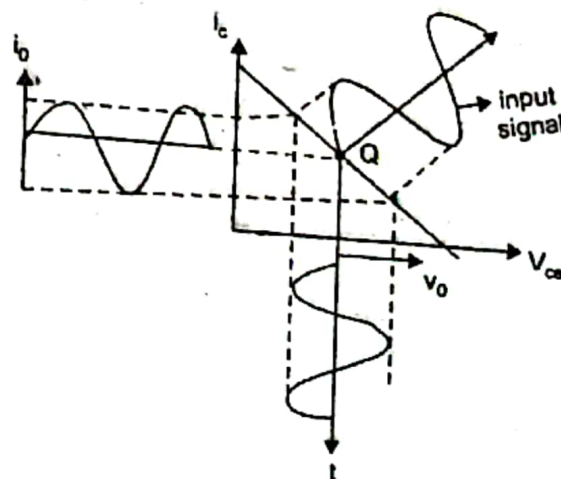
$$f = \frac{1}{2.3 \times 60 \times 10^3 \times 0.25 \times 10^{-6} \log_{10} \frac{1}{1-0.65}}$$

$$f = 63.5 \text{ Hz.}$$

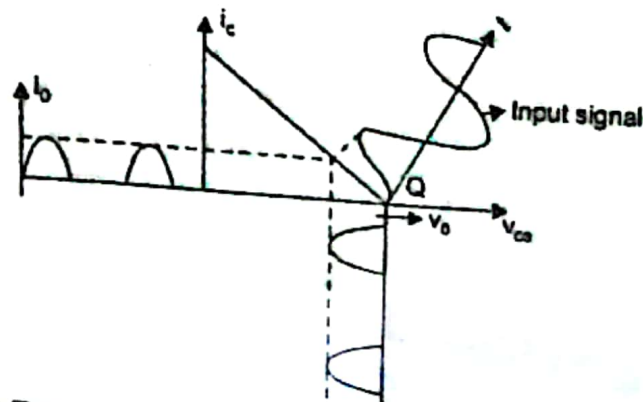
UNIT-IV

Q.8. (a) Write down the classification of a power amplifier. Draw circuit diagram and their characteristics also. (6.5)

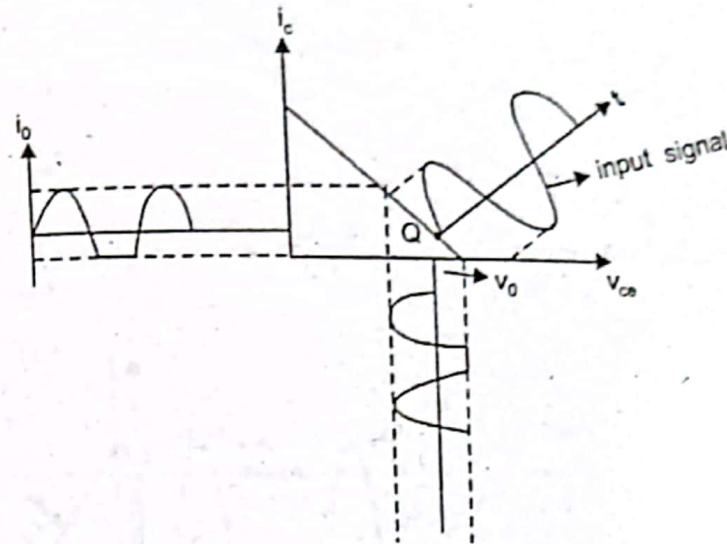
Ans. (i) **Class A Power Amplifier:** When an amplifier is so biased that Q point lies at the centre of load line in active region. Output current flows for all the 360° of the cycle without any clipping or distortion.



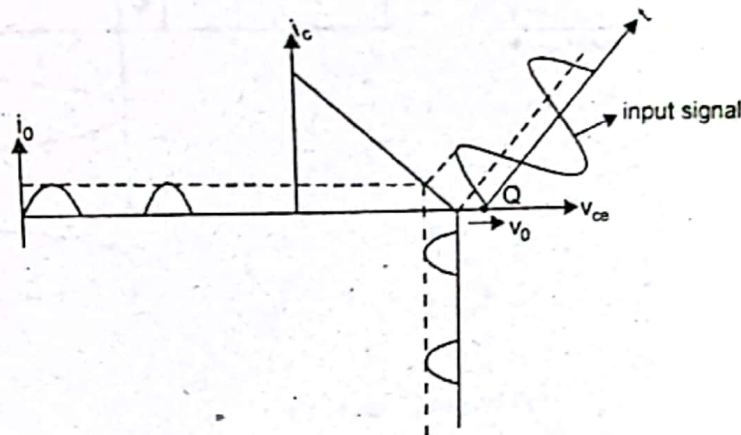
(ii) **Class B Power Amplifier:** When Q point of amplifier is fixed at cut off and output current flows only for 180° (half cycle).



(iii) **Class AB Power amplifier:** When Q point lies just above the cut off on the load line but below the centre point of the load line. Output current flows more than 180° and less than 360° .



(iv) **Class C Power Amplifier:** Q point is fixed below the cut off point then some positive part is lost and output current flows only for less than 180° .



Q.8. (b) Derive the conversion efficiency of a class B power amplifier and clearly explain the reasons, why are most of the power amplifiers used in practice designed to operate in class AB stage? (6)

Ans. Class B Power Amplifier

In class B operation the transistor is so biased that zero signal collector current is zero. Hence class B operation does not need. Any biasing system. The operating point is set at cutoff, as illustrated in Fig.. It remains forward biased for only half cycle of the input signal i.e., its conduction angle is 180° .

As illustrated in Fig., during the positive half cycle of the input ac signal, the circuit is forward biased and, therefore, collector current flows. On the other hand during negative half cycle of the input ac signal, the circuit is reverse biased and no collector current flows.

Power and Efficiency Calculations

Input dc power, $P_{in}(dc) = V_{cc} I_{dc}$ where I_{dc} is the average or direct current taken from the collector supply.

If I_{max} is the maximum or peak value of collector or output current, then

22-2017

Third Semester, Analog Electronics -I

$$I_{dc} = \frac{1}{2\pi} \int_0^\pi I_{c\max} \sin\theta d\theta = \frac{I_{c\max}}{2\pi} [-\cos\theta]_0^\pi = \frac{I_{c\max}}{\pi}$$

$$\therefore i_c = I_{c\max} \sin\theta \text{ for } 0 < \theta < \pi$$

$$\text{RMS value of collector current, } I_{crms} = \frac{I_{c\max}}{\sqrt{2}}$$

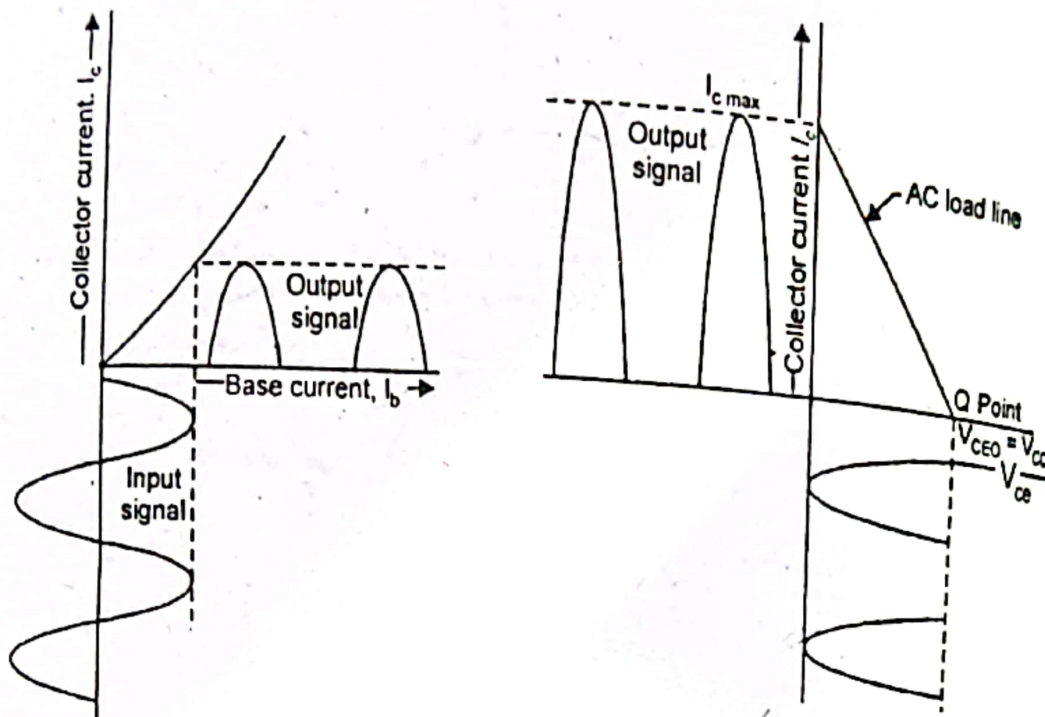


Fig. Class B operation

$$\text{RMS value of output voltage, } V_{rms} = \frac{V_{CC}}{\sqrt{2}}$$

Hence output power during half cycle,

$$P_{out(ac)} = \frac{1}{2} \frac{I_{c\max}}{\sqrt{2}} \times \frac{V_{CC}}{\sqrt{2}} = \frac{I_{c\max} V_{CC}}{4}$$

In above equation factor $\frac{1}{2}$ is used because power is developed during one half cycle only.

$$\text{DC power loss in load, } P_{R_L(dc)} = I_{dc}^2 R_C = \left(\frac{I_{c\max}}{\pi} \right)^2 R_C$$

DC power loss in collector region or transistor

$$= P_{in(dc)} - P_{R_L(dc)} - P_{out(ac)}$$

Overall efficiency,

$$\begin{aligned}
 \frac{P_{out(ac)}}{P_{in(dc)}} &= \frac{I_{c\max} V_{cc} / 4}{I_{c\max} V_{cc} / \pi} \\
 &= \frac{\pi}{4} = 0.785 \text{ or } 78.5\%
 \end{aligned}$$

Conclusions: 1 Severe distortion occurs because of total absence of negative half cycle from the output.

2. As average current in class B operation $\frac{(I_{c\max})}{\pi}$ is less than that in class A operation, so power dissipation is less and consequently overall efficiency is high [78.5% when peak input signal makes $V_{ce\min} = 0$]. In general, overall efficiency is given as

$$\eta_{\text{overall}} = 78.5 \left[1 - \frac{V_{ce\min}}{V_{cc}} \right] \%$$

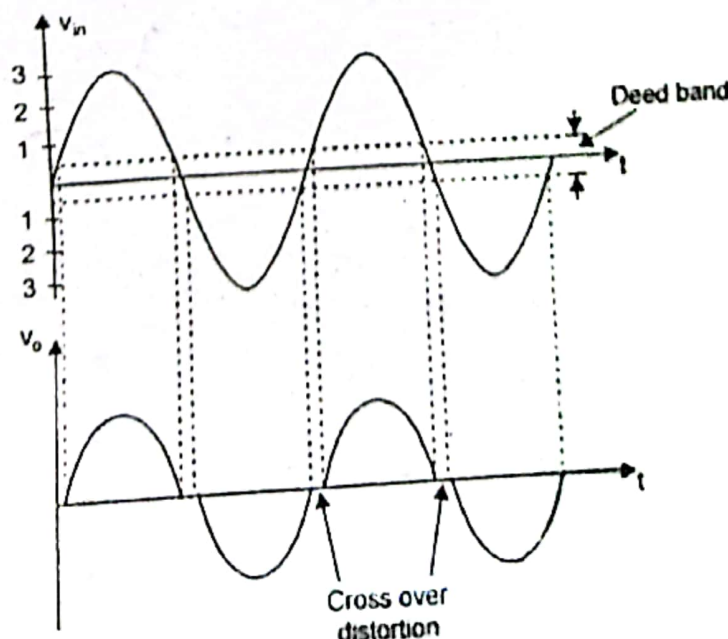
3. Zero signal input represents the best condition for class B operation as collector current is zero. This is quite contrary to class A operation. In class B operation the transistor dissipation increases with creased input signals.

Q.9. Write short note on the following:

(6.25)

Q.9. (a) Cross over distortion.

Ans. The Silicon transistors used in class B push-pull amplifier must have at least 0.5V to 0.6V of forward base-emitter bias before they will go into conduction but the forward bias produced by the input, both the transistor will be non-conducting, when the input signal is approximately ± 0.5 V. This forms a dead band in the input and produces cross-over distortion in the output.



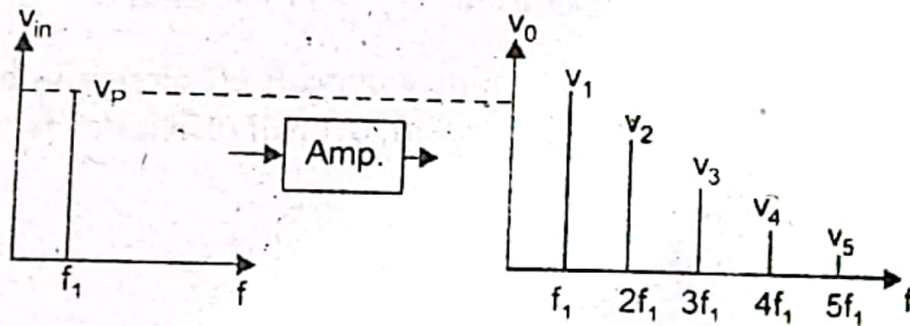
24-2017

Third Semester, Analog Electronics -I

To eliminate crossover distortion, it is necessary to add a small amount of forward bias to take the transistors to average of conduction.

Q.9. (b) Harmonic distortion.

Ans. The frequency domain of non-linear distortion is known as harmonic distortion. (6.25)
The input signal is a single frequency signal (f_1) but the output consist of dc component and different harmonics. These harmonics are integral multiples of the input signal frequency. Thus we obtain distorted output. The distortion magnitude depends on the strength and number of these harmonics. For the better performance of the amplifier, the less harmonic distortion is required. For speech, harmonic distortion should not exceed 10%. High Fidelity or Hi-Fi amplifier have harmonic distortion of less than 1 percent.



END TERM EXAMINATION [DEC-2016] THIRD SEMESTER [B.TECH] ANALOG ELECTRONICS-I [ETEC-203]

Time : 3 hrs.

Note: Attempt any five questions including Q.No.1 which is compulsory. Select one question from each unit. Assume missing data if any.

M.M.: 75

Q.1. Attempt all the questions:

Q.1. (a) What is emitter follower? Explain.

Ans. Emitter follower circuit:

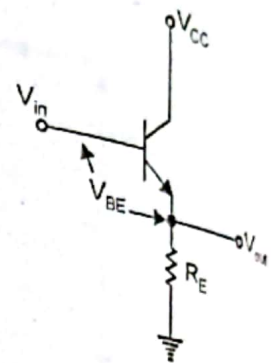
- It is a single transistor in common collector configuration.

- It is also known as Buffer, voltage follower and unity follower due to unity voltage gain ($A_v \leq 1$), current gain depends on $\beta(h_{fe})$.

- Output and Input ac voltage are in same phase and approximately equal in magnitude.

$$V_{out} = V_{in} - V_{BE} \approx V_{in} (\because V_{BE} \ll V_{in})$$

- It has distortionless output and large Bandwidth.
- No loading effect due to high input impedance.



(2.5×10=25)

Q.1. (b) What is the significance of stability factor? Define $S_{(I_{CO})}$, $S_{(\beta_{dc})}$ and $S_{(V_{BE})}$.

Ans. Stability Factor: The stability of Q point of a transistor amplifier depends on the following three parameters:

1. Leakage current I_{CO}

2. β_{dc}

3. Base to emitter voltage V_{BE}

The effect of these parameters can be expressed mathematically by defining the stability factors for the three parameters individually as follows:

(i) Stability factor, $S = \frac{\Delta I_C}{\Delta I_{CO}}$ Constant V_{BE} and β_{dc} or $\frac{\partial I_C}{\partial I_{CO}}$

This represents the change in collector current due to change in reverse saturation current I_{CO} . The other two parameters that means V_{BE} and β_{dc} are assumed to be constant.

(ii) Stability factor, $S' = \frac{\Delta I_C}{\Delta V_{BE}}$ Constant I_{CO} and β_{dc} or $\frac{\partial I_C}{\partial V_{BE}}$

S' represents the change in I_C due to change in V_{BE} at constant I_{CO} and β_{dc}

(iii) Stability factor, $S'' = \frac{\Delta I_C}{\Delta \beta_{dc}}$ Constant I_{CO} and V_{BE} or $\frac{\partial I_C}{\partial \beta_{dc}}$

These expression indicate that I_C is collectively dependent on I_{CO} , V_{BE} and β_{dc} . The combined effect to these parameters on I_C is mathematically given by,

Total change in collector current

$$\Delta I_C = S \Delta I_{CO} + S' \Delta V_{BE} + S'' \Delta \beta_{dc}$$

Ideally the values of all the stability factors should be zero and practically they should be as small as possible.
Practically the value of S is significantly higher than the other two stability factors. Hence while comparing the biasing circuits, we should focus our attention more on the value of S .

Q.1. (c) Explain the function of coupling capacitor and bypass capacitor?

Ans. The coupling capacitor C_c transmits ac signal but blocks the dc voltage of the first stage from reaching the base of the second stage. Thus the dc biasing of the next stage is not interfered.

The Emitter bypass capacitor C_E , offers low reactance path to the signal. If it is not present, then the voltage drop across R_E will reduce the effective voltage available across the base-emitter terminals and thus reduce the gain.

Q.1. (d) Why the channel of a JFET is never completely closed at the drain end?

Ans. The Channel is not completely closed at the drain end, because in that case there will be no drain current, so there will be no voltage drop along the channel length and amount of reverse bias will become uniform and the wedge-shaped depletion region will become rectangular one.

Q.1. (e) Define "current controlled device" and "voltage controlled device"?

Ans. A current controlled device is one in which current defines the operating conditions of the device, whereas a voltage controlled device is one in which a particular voltage defines the operating conditions.

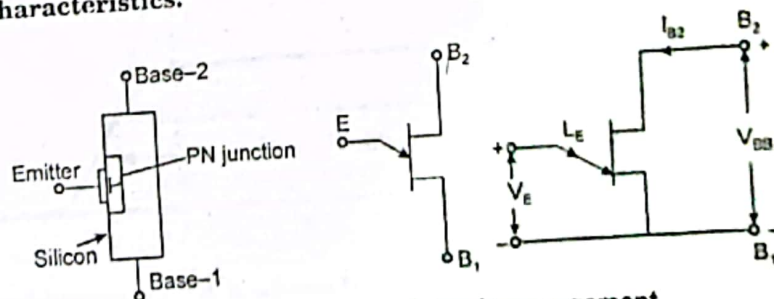
Q.1. (f) Why FET is called a voltage controlled device?

Ans. In a FET, drain current is controlled by the effect of the extension of the field associated with the depletion region developed by the reverse bias on the gate, so it is called a voltage controlled device.

Q.1. (g) What is negative resistance region in UJT? Explain its physical significance.

Ans. UJT (Unijunction Transistor): It is also called the double base diode is a 2 layer 3 terminal solid state (silicon) switching device. The device has a unique characteristic that when it is triggered, its I_E increases regeneratively until it is restricted by emitter power supply.

V-I characteristics:



Basic structure, Symbol, Basic arrangement

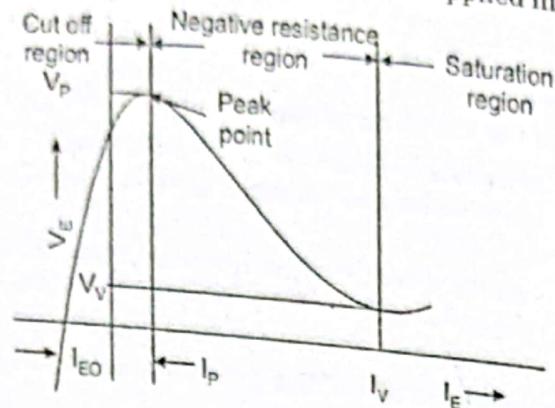
The device has only one Junction, so it is called the unijunction device. Its structure is similar to N-channel JFET.

UJT is operated with Emitter J^n FB while the JFET is normally operated with gate J^n RB.

It exhibits a negative resistance characteristic and so it can be employed as an oscillator. It has high pulse current capability.

A very low value of triggering current lost cost.

A stable triggering voltage (V_P)-a fixed fraction of applied inter base voltage V_{BB} .



Emitter Characteristic of a UJT

Q.1. (h) How does negative feedback increases bandwidth of an amplifier?

Ans. The gain with negative feedback $V_{vf} = \frac{A_{v0}}{1 + A_{v0}\beta}$

Where A_{v0} is gain without feedback.

The gain decreases with introduction of negative feedback. The amplifiers exhibit a constant gain-bandwidth product. The reduction in gain is therefore, accompanied by increase in bandwidth. The bandwidth increases by the same factor $(1 + A_{v0}\beta)$ by which the gain reduces as shown in the figure.

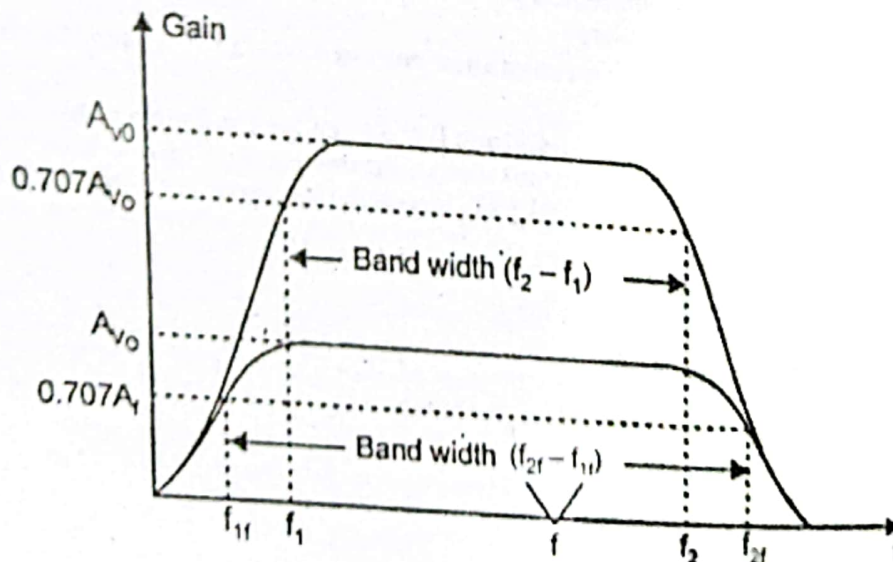


Fig. Voltage gain is decreased and bandwidth is increased for an amplifier using negative feedback.

Q.1. (i) What advantages does the transformer provide in a transformer coupled power amplifier?

Ans. Transformer provides the following advantages in a transformer coupled power amplifier.

1. Power loss in the primary of the transformer used for coupling the load to the amplifier, is negligible and therefore conversion efficiency is doubled in a transformer coupled power amplifier.

2. Output transformer makes the transfer of maximum power to the load through impedance matching.

3. Output transformer prevents a large dc current from flowing in the load. Which otherwise could be harmful if the load were a loudspeaker.

Q.1. (j) What are the drawbacks of complementary symmetry push pull class B power amplifier?

Ans. One obvious drawback of this circuit arrangement is the requirement for two supply voltages. Another drawback is difficult of obtaining matched complementary transistors. If there is an unbalance in the characteristics of the two transistors, even harmonic will no longer be cancelled and considerable distortion will be introduced.

UNIT-I

Q.2. (a) A Si transistor has a potential divider biasing. If it operates at $I_C = 3\text{mA}$ and $V_{CE} = 10\text{V}$. Find V_{CC} , R_{B1} , V_B and R_E for $\beta = 100$. Assume $V_{BE} = 0.7\text{V}$, $R_{B2} = 20\text{K}\Omega$ and $R_C = 5\text{K}\Omega$. (6)

Ans. $I_C = \beta I_B$

$$I_B = \frac{I_C}{\beta} = \frac{3 \times 10^{-3}}{100} = 0.03\text{mA}$$

$$I_1 = 10 I_B = 10 \times 0.03 = 0.3\text{mA}$$

The value of R_{B1} is so decided that the current I flowing through R_{B1} and R_{B2} ($20\text{K}\Omega$) is about 10 times of the base current to achieve good stabilization by the circuit shown

$$V_{CB} + V_{BE} = V_{CE}$$

$$V_{CB} = V_{CE} - V_{BE} = 10 - 0.7$$

$$= 9.3\text{ Volt}$$

and $I_1 R_{B1} = I_C R_C + V_{CB}$

$$= 3 \times 10^{-3} \times 5 \times 10^3 + 9.3$$

$$= 24.3\text{ V}$$

$$R_{B1} = \frac{24.3}{I_1} = \frac{24.3}{0.3 \times 10^{-3}}$$

$$R_{B1} = 81\text{ K}\Omega$$

$$V_B = I_1 \times R_{B2} = 0.3 \times 10^{-3} \times 20 \times 10^3$$

$$V_B = 6\text{ Volt}$$

$$V_{CC} = I_1 (R_{B1} + R_{B2})$$

$$= 0.3 \times 10^{-3} (81 + 20) \times 10^3$$

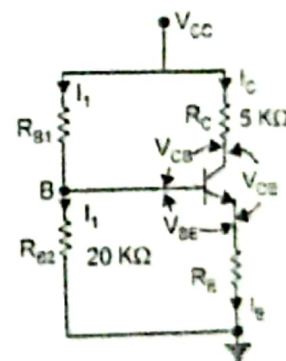
$$V_{CC} = 30.3\text{ Volt}$$

Now $I_E R_E = V_B - V_{BE}$

$$I_E R_E = 6 - 0.7 = 5.3\text{ Volt}$$

$$R_E = \frac{5.3}{I_E} = \frac{5.3}{3 \times 10^{-3}}$$

$$R_E = 1.77\text{ K}\Omega \quad (\text{Assuming } I_C = I_E)$$



Q.2. (b) Explain how phase reversal of the signal takes place when it is amplified by a single stage CE voltage amplifier? (2.5)

Ans. In a CE configuration, the output voltage increases in the negative direction when the input signal voltage increases in the positive direction and vice-versa. This is called the phase reversal and causes a phase difference of 180° between the input signal voltage and output voltage.

Q.2. (c) How many types of biasing are done on a BJT to work properly as an amplifier? Which one is the best and why?

Ans. Following are the types of biasing technique used on a BJT.

- (i) Fixed bias
- (ii) Collector to base bias
- (iii) Fixed bias with emitter resistor
- (iv) Voltage divider or self bias circuit.

Voltage divider or self bias circuit is the most commonly used biasing arrangement. Its popularity is primarily due to its low sensitivity to variations in β from one transistor to another of the same lot.

Q.3. (a) Derive expression for voltage gain, current gain, input resistance and output resistance of a common base Si-transistor circuit using h parameters.

Ans.

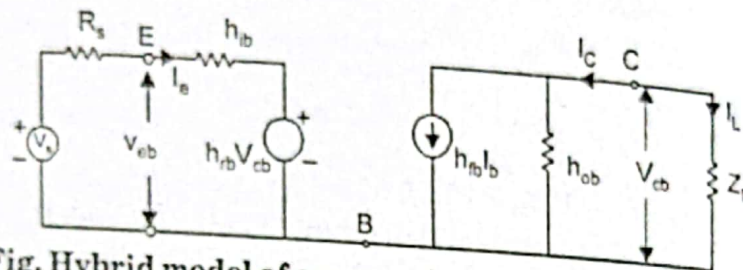


Fig. Hybrid model of common base transistor amplifier

Current gain

$$A_I = \frac{I_L}{I_e} = -\frac{I_C}{I_e}$$

But

$$I_C = h_{fb} I_b + h_{ob} V_{cb}$$

or

$$I_C = h_{fb} I_b + h_{ob} (-I_C Z_L)$$

$$I_C = h_{fb} I_b - h_{ob} h_{fb} I_b Z_L$$

$$I_C = h_{fb} I_b (1 - h_{ob} Z_L)$$

and

$$I_e = I_C + I_b = h_{fb} I_b + I_b$$

$$I_e = I_b (h_{fb} + 1)$$

$$(\because I_e = h_{fb} I_b)$$

$\therefore$

$$A_I = \frac{I_C}{I_e} = \frac{h_{fb} I_b (1 - h_{ob} Z_L)}{I_b (h_{fb} + 1)}$$

$$A_I = \frac{h_{fb} (1 - h_{ob} Z_L)}{(h_{fb} + 1)}$$

The input impedance (Z_i)

$$A_I \approx (1 - h_{ob} Z_L)$$

$$(\because h_{fb} \gg 1)$$

$$Z_i = \frac{V_{eb}}{I_b}$$

$\therefore$

$$V_{eb} = h_{ib} I_e + h_{rb} V_{cb}$$

$$= h_{ib} (h_{fb} + 1) I_b + h_{rb} h_{fb} I_b Z_L$$

$$\frac{V_{eb}}{I_b} = h_{ib} (h_{fb} + 1) + h_{rb} h_{fb} Z_L$$

$$(\because h_{fb} \gg 1)$$

$$Z_i = h_{ib} (h_{fb} + 1) + h_{rb} h_{fb} Z_L$$

Voltage Gain (A_v)

$$A_v = \frac{V_{cb}}{V_{eb}} = \frac{h_{fb} I_b Z_L}{h_{ib} h_{fb}}$$

$$A_v = \frac{I_b Z_L}{h_{ib}}$$

Output Impedance (Z_o)

$$Z_o = \frac{V_{cb}}{I_c} = \frac{-I_c Z_L}{I_c}$$

$$Z_o = Z_L$$

Q.3. (b) What is Darlington pair? Derive an expression for current gain in Darlington pair? (5)

Ans. Refer to Q. 1. (c) First Term 2016

UNIT-II

Q.4. (a) Draw the cascade of a common emitter stage and common collector and give the expression for input resistance and current gain of the circuit. (7.5)

Ans.

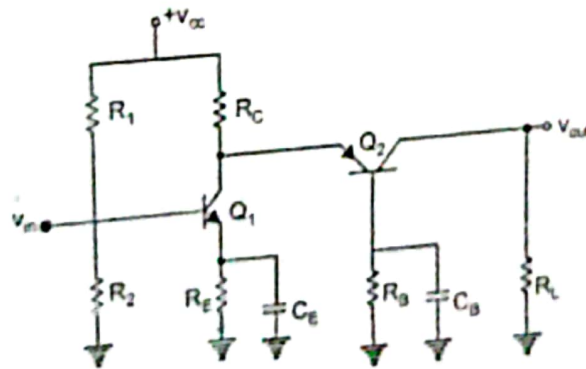


Fig. Cascade Amplifier

This arrangement is designed to provide a high input impedance with low voltage gain to ensure that input miller capacitance is at a minimum with CB stage providing high frequency operation.

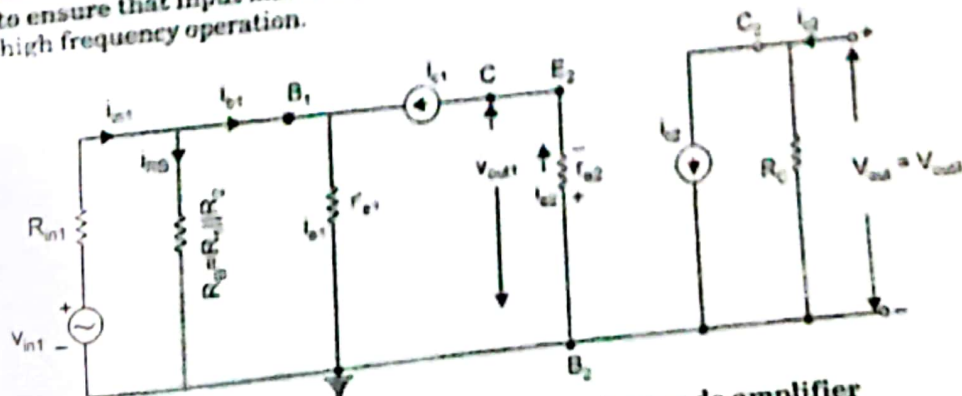


Fig. Small signal T-equivalent circuit of cascade amplifier

Current Gain:

The current gain A_i of cascade (CE-CB) configuration may be expressed as

$$A_i = \frac{i_{out}}{i_{in1}} = \frac{i_{b1}}{i_{in1}} \times \frac{i_{e1}}{i_{b1}} \times \frac{i_{e2}}{i_{e1}} \times \frac{i_{c2}}{i_{e2}} \times \frac{i_{out}}{i_{c2}}$$

$$= \frac{i_{b1}}{i_{in1}} \times \frac{i_{e1}}{i_{b1}}$$

Applying current divider rule to the input circuit

$$i_{b1} = \frac{R_B i_{in1}}{R_B + \beta_{ac} r_{e1}}$$

Also

$$i_{e1} = \beta_{ac} i_{b1}$$

$$\therefore A_i = \frac{\beta_{ac} R_B}{R_B + \beta_{ac} r_{e1}}$$

Input resistance:

$$V_{in1} = r_{e1} i_{e1} = \beta_{ac} r_{e1} i_{b1}$$

$$V_{in1} = \beta_{ac} r_{e1} \times \frac{R_B i_{in1}}{R_B + \beta_{ac} r_{e1}}$$

$$\text{and input resistance } R_{in} = \frac{V_{in1}}{i_{in1}}$$

$$R_{in} = \frac{\beta_{ac} r_{e1} R_B}{R_B + \beta_{ac} r_{e1}}$$

Q.4. (b) A single stage amplifier has a gain of 60. The collector load $R_C = 500 \Omega$ and the input impedance is $1 K\Omega$, calculate the overall voltage gain when two stages are cascaded through RC coupling. (5)

Ans. In case of mid-frequency range the overall voltage gain

$$A_{vm} = \frac{h_{fe} R_C}{R_C + h_{ie}}$$

$$A_{vm} = \frac{60 \times 500}{500 + 1000}$$

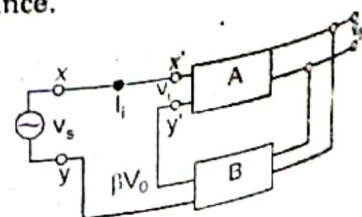
$$= \frac{30000}{1500}$$

$$A_{vm} = 20.$$

Q.5. (a) How the negative feedback changes the impedances? Explain with mathematical expression? (7.5)

Ans. Effect of negative feedback on input resistance.

(1) **Series Feedback:** Change in Input Impedance (Voltage Series Feedback). Due to the application of negative feedback, the input impedance increases. High input impedance is desirable in an amplifier because it will not load the input voltage source. The effect of negative feedback on Input impedance can be considered.



Let v_s be the Input voltage applied to the amplifier at terminals xy and v_i be the Input Voltage at the Input terminals is $X'Y'$.
 Suppose the input impedance of the amplifier is Z_i without feedback and Z_{if} the input impedance with negative feedback. If I_i be the Input current then

$$Z_{if} = V_s / I_i$$

$$Z_i = V_i / I_i$$

$$V_i = V_s - BV_o$$

$$V_o = Av_i$$

$$v_i = V_s - BA v_i$$

$$v_i(1 + AB) = V_s$$

$$\frac{V_s}{I_i}(1 + AB) = Z_{if}$$

$$Z_{if} = Z_i(1 + AB)$$

Z_{if} increased by the factor $(1 + AB)$

(2) Voltage shunt feedback:

$$I_s = I_{in} + I_f$$

input impedance

$$Z_{inf} = \frac{V_{in}}{I_s} = \frac{V_{in}}{I_{in} + I_f}$$

$$= \frac{V_{in}}{I_{in} + \beta V_{out}}$$

$$= \frac{V_{in} / I_{in}}{1 + \frac{\beta V_{out}}{I_{in}}}$$

$$Z_{inf} = \frac{Z_{in}}{1 + A\beta} \quad [Z_{in} \text{ decreased by the factor } (1 + A\beta)]$$

Current-shunt Feedback

$$I_s = I_{in} + I_f = I_{in} + \beta I_{out}$$

$$I_s = I_{in}(1 + A\beta)$$

$$Z_{inf} = \frac{V_{in}}{I_s} = \frac{V_{in}}{I_{in}(1 + A\beta)}$$

$$Z_{inf} = \frac{V_{in} / I_{in}}{(1 + A\beta)}$$

$$Z_{inf} = \frac{Z_{in}}{1 + A\beta} \quad [Z_{in} \text{ is decreased by the factor } (1 + A\beta)]$$

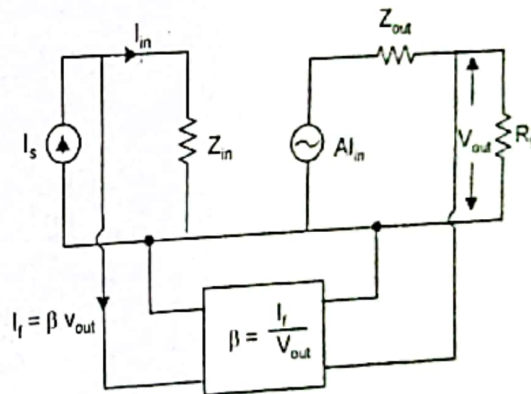
Effect of negative feedback on output impedance:

(1) Voltage series feedback:

$$V_{out} = I_{out} Z_{out} + AV_{in}$$

$$V_{out} = I_{out} Z_{out} - AV_f$$

$$(\because V_{in} = -V_f)$$



$$V_{out} = I_{out} Z_{out} - A\beta V_{out} \quad \text{and } V_f = \beta V_{out}$$

$$V_{out}(1 + A\beta) = I_{out} Z_{out}$$

$$Z_{outf} = \frac{V_{out}}{I_{out}} = \frac{Z_{out}}{(1 + A\beta)}$$

Z_{outf} is decreased by the factor $(1 + A\beta)$

Current shunt feedback:

with

$$I_s = 0$$

$$I_{in} = -I_f = -\beta I_{out}$$

$$= -\beta I$$

$$\therefore I = \frac{V}{Z_{out}} - A I_f$$

$$= \frac{V}{Z_{out}} - A\beta I$$

$$I(1 + A\beta) = \frac{V}{Z_{out}}$$

$$Z_{outf} = \frac{V}{I} = Z_{out}(1 + A\beta)$$

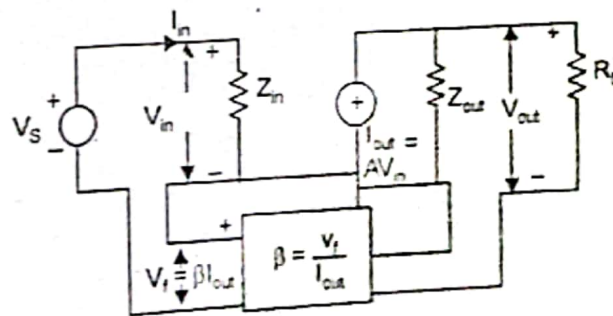
– Output impedance is increased by the factor $(1 + A\beta)$

Current Series feedback

$$I = \frac{V}{Z_{out}} - A V_{in} = \frac{V}{Z_{out}} - A V_f$$

$$I = \frac{V}{Z_{out}} - A\beta I_{out} \quad [\because V_{in} = V_f = \beta I_{out} \text{ and } I_{out} = I]$$

$$I(1 + A\beta) = \frac{V}{Z_{out}}$$



$$Z_{outf} = \frac{V}{I} = Z_{out}(1 + A\beta)$$

– Output impedance is increased by the factor $(1 + A\beta)$

5. (b) Why voltage series feedback is most commonly used in cascaded amplifier?

Ans. In cascaded amplifiers, high input impedance and low output impedance are the main requirements and both these requirements can be met by voltage feedback, so it is most commonly used in cascaded amplifiers. (2)

5. (c) The open loop gain of an amplifier changes by 5%. If 10 dB negative feedback is applied, calculate the percentage change in the closed loop gain.

Ans. Percentage change in closed loop gain (3)

$$\frac{dA_f}{A_f} = \frac{dA}{A} \frac{1}{1 + A_\beta} \times 100$$

here $\frac{dA}{A} = 0.05$

and $A_\beta = 10\text{dB} = 3.16$

$$\begin{aligned} \frac{dA_f}{A_f} &= \frac{0.05}{1 + 3.16} \times 100 = \frac{0.05}{4.16} \times 100 \\ &= 1.2\% \end{aligned}$$

UNIT-III

6. (a) Explain the meaning of pinch off voltage in a JFET. How does the current flow in a JFET after pinch off? Also explain the meaning of the terminologies: (6)

i) Trans-conductance

ii) Drain Resistance and

iii) Amplification Factor.

Ans. **Pinch off voltage:** The value of drain-source voltage (V_{ds}) at which channel is pinched-off (All the free charges from the channel get removed) is called pinch-off voltage.

The amount of reverse bias is not the same throughout the length of the p-n junction, the channel becomes narrower at the drain end in comparison to that at source. The channel is not completely closed at the drain end. Because in that case there will be no drain current, so there will be no voltage drop along the channel length and the amount of reverse bias will become uniform and wedge-shaped depletion region will be rectangular one. The channel will open and the drain current will flow. However, at pinch-off voltage, the channel width is reduced to a constant minimum value to allow flow of drain current.

Dynamic Drain Resistance (r_d): Drain resistance is an AC resistance of JFET. It is defined as the ratio of change in the drain to source voltage to the corresponding change in the drain current, at a constant value of gate to source voltage. Drain resistance is calculated in the saturation region of the FET output characteristics.

$$\frac{\Delta V_{DS}}{\Delta I_D} \bigg|_{V_{GS} \text{ is constant}}$$

Trans conductance (g_m): Trans conductance g_m is defined as the ratio of change in drain current to the corresponding change in gate to source voltage, at a constant value of drain to source voltage.

$$g_m = \left. \frac{\Delta I_D}{\Delta V_{GS}} \right|_{V_{DS} \text{ is constant}}$$

Amplification factor (μ): Amplification factor ' μ ' is defined as the ratio of change in the drain to source voltage, to change in the gate to source voltage, at a constant value of drain current I_D .

$$\mu = \left. \frac{\Delta V_{DS}}{\Delta V_{GS}} \right|_{I_D \text{ is constant}}$$

Relation between r_d , g_m and μ :

$$\mu = \frac{\Delta V_{DS}}{\Delta V_{GS}}$$

or

$$\mu = \frac{\Delta V_{DS}}{\Delta I_D} \times \frac{\Delta I_D}{\Delta V_{GS}}$$

$$\boxed{\mu = r_d \times g_m}$$

Q.6. (b) Calculate the minimum value of V_{DS} required for an NMOSFET to operate in pinch off condition when $V_{GS} = 1V$, $V_p = -2V$, $I_{DSS} = 10 \text{ mA}$. Also find the corresponding drain current.

(4)

Ans.

$$I_D = I_{DSS} \left[1 - \frac{V_{GS}}{V_p} \right]^2$$

$$= 10 \times 10^{-3} \left[1 + \frac{1}{2} \right]^2$$

$$= 10 \times 10^{-3} \left[\frac{3}{2} \right]^2 = 10 \times 10^{-3} \times \frac{9}{4}$$

$$I_D = 22.5 \times 10^{-3} \text{ A}$$

$$I_D = 22.5 \text{ mA}$$

(2.5)

Q.6. (c) Give a comparison between JFET and MOSFET.

Ans. JFET and MOSFETs are quite similar in their operating principles and their electrical characteristics. However, they differ in some aspects, as detailed below

1. In MOSFETs (D-MOSFETs as well as E-MOSFETs), the transverse electric field induced across an insulating layer deposited on the semiconductor material controls the conductivity of the channel whereas in the JFETs the transverse electric field across the reverse biased PN junction controls the conductivity of the channel

2. JFETs can only be operated in depletion mode whereas MOSFETs can be operated in their depletion or in enhancement mode

3. MOSFETs have input impedance much higher than that of JFETs because of small leakage current.
4. JFETs have characteristics curves more flatter than those of MOSFETs indicating higher drain resistance.
5. MOSFET is very susceptible to overload voltage and needs special handling during installation
6. Capacitive effects may be considerably lower in dual gate MOSFETs.
7. Special digital CMOS circuits are available which involve near zero power dissipation and very low voltage and current requirements.
8. When JFET is operated with a reverse bias on the junction the gate current I_G is larger than it would be in a comparable MOSFET
9. MOSFETs are some what easier to manufacture, they are more widely used than the JFETs.

Q.7. (a) Sketch and explain the silicon controlled Rectifier characteristics with the switching action.

(7.5)

Ans. SCR: Silicon Controlled Rectifier is constructed of a semiconductor material with a third terminal for control purpose.

Silicon was chosen because of its high temperature and power capabilities. It is a 3 terminal device, Anode, cathode and gate. Gate determines when the rectifier swides from the open circuit to short circuit gate. It is not enough simply to forward bias the anode to cathode region of the device. The dynamic resistance of the SCR is typically 0.01 to 0.1 Ω and reverse resistance is typically 100 K Ω or were.

It is widely used as a switching devices is power control applications. It can control leads by switching *ON* and *OFF* upto many thousand times a second. It can switch *ON* for a variable length of time duration, thereby delivering desired amount of power to the load. Thus, it possesses the advantage of a cheostat as well as a switch with have of their drawback.

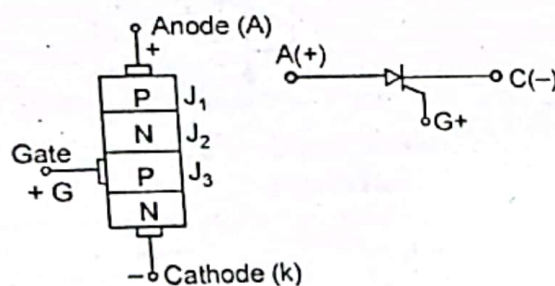


Fig. Symbol of SCR

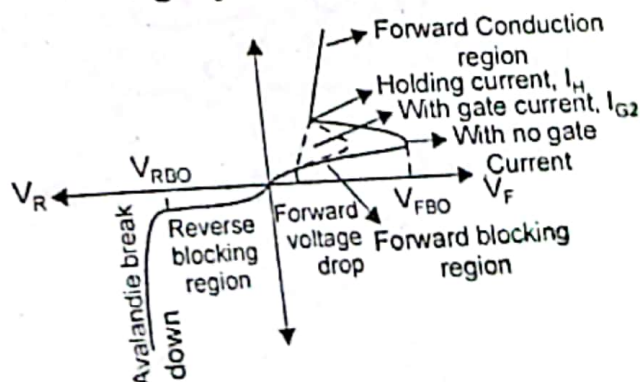


Fig. Characteristics

Q.7. (b) A UJT relaxation oscillator has $R_E = 60\text{ K}\Omega$ and $C = 0.25\text{ }\mu\text{F}$. Determine the pulse repetition frequency. Take intrinsic stand-off ratio to be 0.65? (5)

Ans. Pulse repetition frequency $f = \frac{1}{2.3R_E C \log_{10} \frac{1}{1-\eta}}$

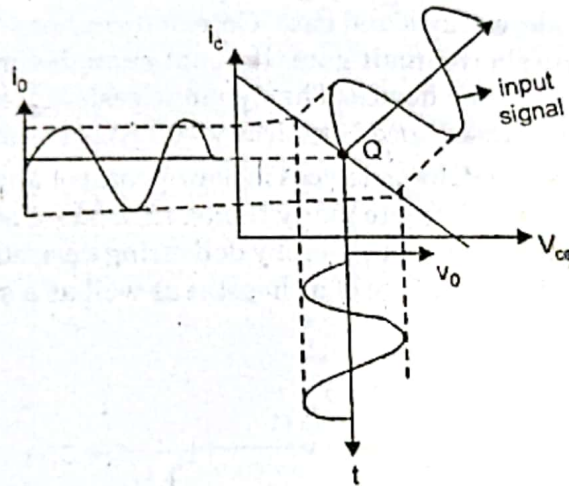
$$f = \frac{1}{2.3 \times 60 \times 10^3 \times 0.25 \times 10^{-6} \log_{10} \frac{1}{1-0.65}}$$

$$f = 63.5\text{ Hz.}$$

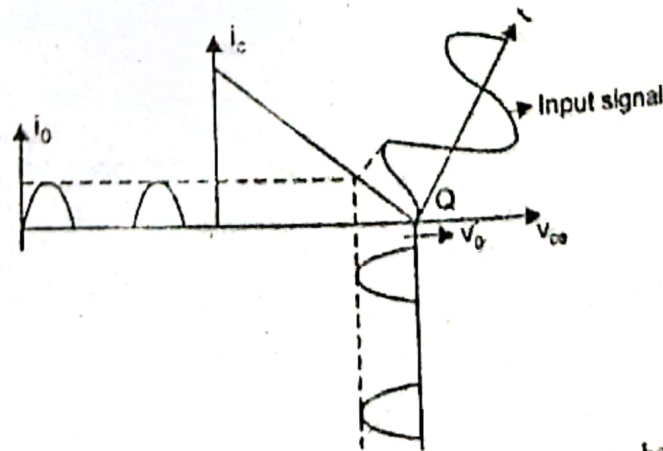
UNIT-IV

Q.8. (a) Write down the classification of a power amplifier. Draw circuit diagram and their characteristics also. (8.5)

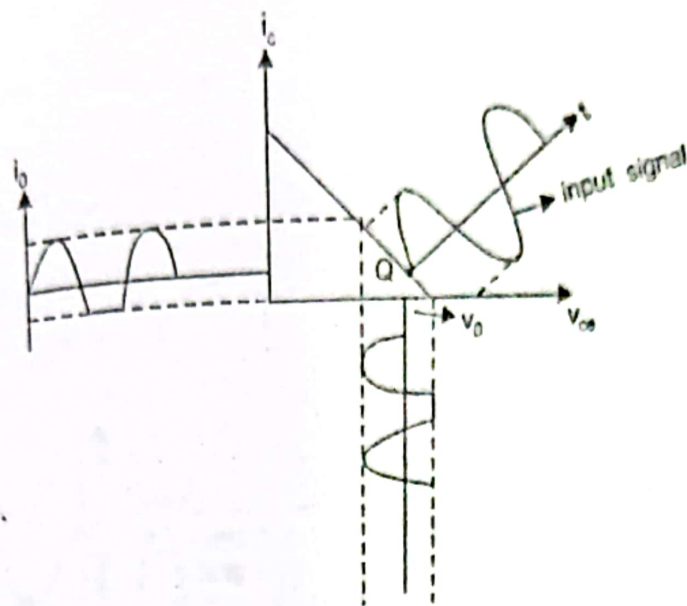
Ans. (i) **Class A Power Amplifier:** When an amplifier is so biased that Q point lies at the centre of load line in active region. Output current flows for all the 360° of the cycle without any clipping or distortion.



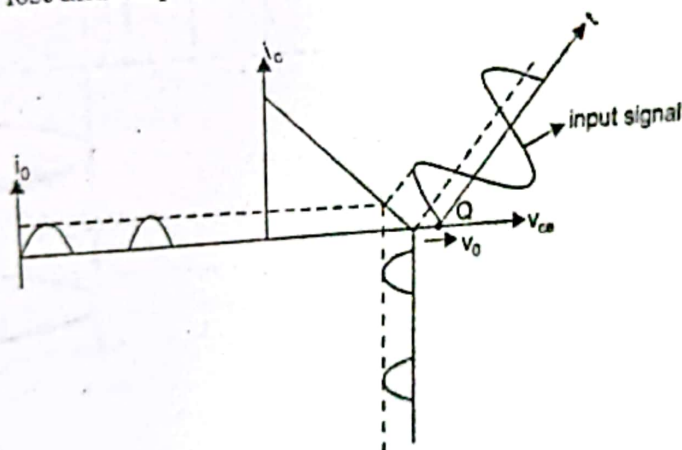
(ii) **Class B Power Amplifier:** When Q point of amplifier is fixed at cut off and output current flows only for 180° (half cycle).



(iii) **Class AB Power amplifier:** When Q point lies just above the cut off on the load line but below the centre point of the load line. Output current flows more than 180° and less than 360° .



(iv) **Class C Power Amplifier:** Q point is fixed below the cut off point then some positive part is lost and output current flows only for less than 180° .



Q.8. (b) Derive the conversion efficiency of a class B power amplifier and clearly explain the reasons, why are most of the power amplifiers used in practice designed to operate in class AB stage? (6)

Ans. Class B Power Amplifier

In class B operation the transistor is so biased that zero signal collector current is zero. Hence class B operation does not need. Any biasing system. The operating point is set at cutoff, as illustrated in Fig.. It remains forward biased for only half cycle of the input signal i.e., its conduction angle is 180° .

As illustrated in Fig., during the positive half cycle of the input ac signal, the circuit is forward biased and, therefore, collector current flows. On the other hand during negative half cycle of the input ac signal, the circuit is reverse biased and no collector current flows.

Power and Efficiency Calculations

Input dc power, $P_{in}(dc) = V_{ce} I_{dc}$ where I_{dc} is the average or direct current taken from the collector supply.

If I_{max} is the maximum or peak value of collector or output current, then

$$I_{dc} = \frac{1}{2\pi} \int_0^{\pi} I_{c \max} \sin \theta d\theta = \frac{I_{c \max}}{2\pi} [-\cos \theta]_0^{\pi} = \frac{I_{c \max}}{\pi}$$

$$\therefore i_c = I_{c \max} \sin \theta \text{ for } 0 < \theta < \pi$$

$$\text{RMS value of collector current, } I_{crms} = \frac{I_{c \max}}{\sqrt{2}}$$

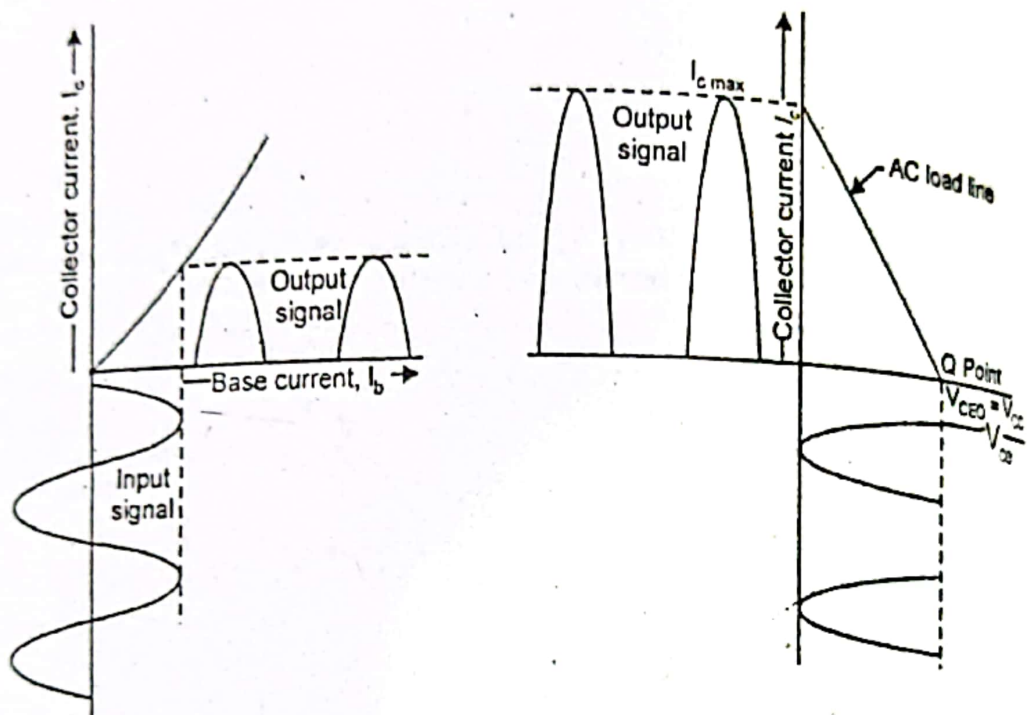


Fig. Class B operation

$$\text{RMS value of output voltage, } V_{rms} = \frac{V_{CC}}{\sqrt{2}}$$

Hence output power during half cycle,

$$P_{out(ac)} = \frac{1}{2} \frac{I_{c \max}}{\sqrt{2}} \times \frac{V_{CC}}{\sqrt{2}} = \frac{I_{c \max} V_{CC}}{4}$$

In above equation factor $\frac{1}{2}$ is used because power is developed during one half cycle only.

$$\text{DC power loss in load, } P_{R_c(dc)} = I_{dc}^2 R_c = \left(\frac{I_{c \max}}{\pi} \right)^2 R_c$$

DC power loss in collector region or transistor

$$= P_{in(dc)} - P_{R_c(dc)} - P_{out(ac)}$$

Overall efficiency,

$$\eta_{\text{overall}} = \frac{P_{\text{out(ac)}}}{P_{\text{in(dc)}}} = \frac{I_{c \text{ max}} V_{CC} / 4}{I_{c \text{ max}} V_{CC} / \pi}$$

$$= \frac{\pi}{4} = 0.785 \text{ or } 78.5\%$$

Conclusions: 1 Severe distortion occurs because of total absence of negative half cycle from the output.

2. As average current in class B operation $\frac{(I_{c \text{ max}})}{\pi}$ is less than that in class A operation, power dissipation is less and consequently overall efficiency is high [78.5% when peak input signal makes $V_{ce \text{ min}} = 0$]. In general, overall efficiency is given as

$$\eta_{\text{overall}} = 78.5 \left[1 - \frac{V_{ce \text{ min}}}{V_{CC}} \right] \%$$

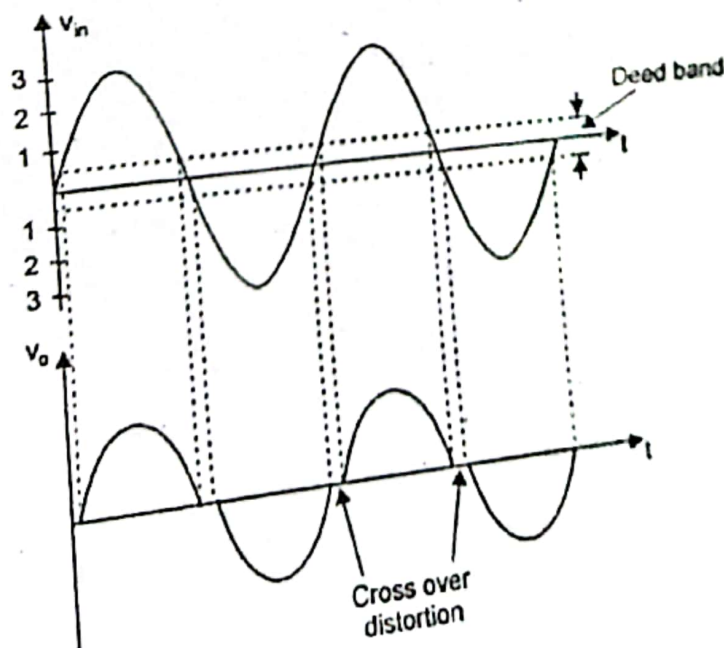
3. Zero signal input represents the best condition for class B operation as collector current is zero. This is quite contrary to class A operation. In class B operation the transistor dissipation increases with creased input signals.

Q.9. Write short note on the following:

(6.25)

Q.9. (a) Cross over distortion.

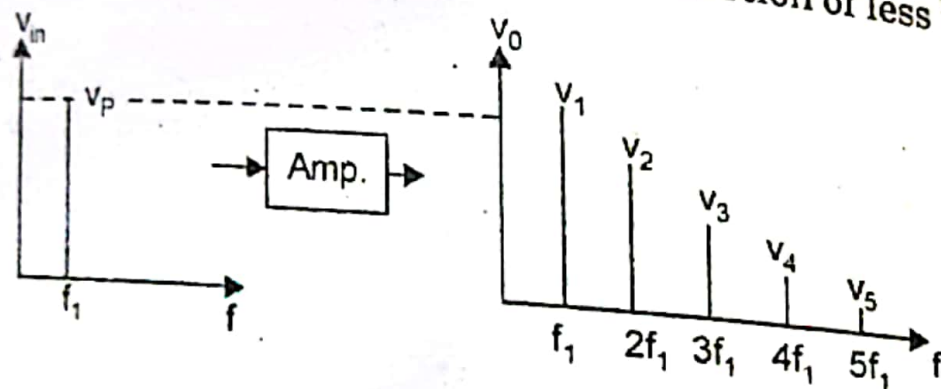
Ans. The Silicon transistors used in class B push-pull amplifier must have at least 0.5V to 0.6V of forward base-emitter bias before they will go into conduction but the forward bias produced by the input, both the transistor will be non-conducting, when the input signal is approximately ± 0.5 V. This forms a dead band in the input and produces cross-over distortion in the output.



To eliminate crossover distortion, it is necessary to add a small amount of forward bias to take the transistors to average of conduction.

Q.9. (b) Harmonic distortion.

Ans. The frequency domain of non-linear distortion is known as harmonic distortion (6.25). The input signal is a single frequency signal (f_1) but the output consists of dc component and different harmonics. These harmonics are integral multiples of the input signal frequency. Thus we obtain distorted output. The distortion magnitude depends on the strength and number of these harmonics. For the better performance of the amplifier, the less harmonic distortion is required. For speech, harmonic distortion should not exceed 10%. High Fidelity or Hi-Fi amplifier have harmonic distortion of less than 1 percent.



END TERM EXAMINATION [DEC. 2015] THIRD SEMESTER [B. TECH] ANALOG ELECTRONICS-I [ETEC-203]

Time: 3 Hrs.

MM : 75

Note: Attempt any five question including Q no.1 which is compulsory. Select one question from each unit.

Q.1. Write short notes on any five of the following:-

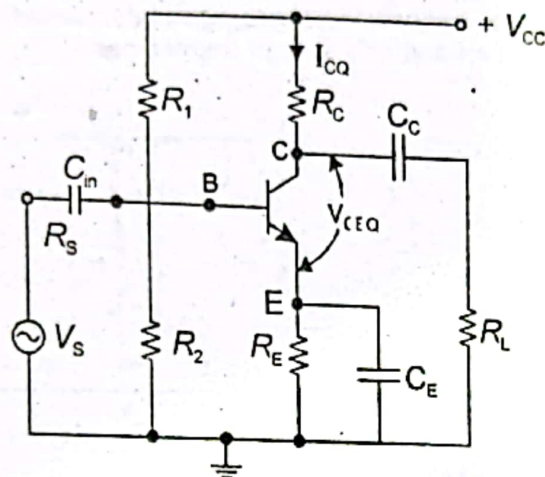
(a) Power dissipation in power amplifiers.

(2.5)

Ans. A power amplifier is required to deliver a large amount of power and as such it has to handle large current. Collector resistance is made low to reduce the power losses.

Let us consider a voltage divider biased common Emitter voltage amplifier circuit.

$$V_{CEQ} = V_{CC} - I_{CQ} (R_C + R_E) \quad \dots(1)$$



DC power supplied to amplifier

$$= V_{CC} I_{CQ}$$

DC power input to transistor

$$= V_{CEQ} I_{CQ}$$

∴

$$\text{Power wasted} = V_{CC} I_{CQ} - V_{CEQ} I_{CQ}$$

$$= I_{CQ} [V_{CC} - V_{CEQ}]$$

$$= I_{CQ} [I_{CQ} (R_C + R_E)] \dots \dots \dots \text{from equation (1)}$$

$$= I_{CQ}^2 (R_C + R_E)$$

Q.1. (b) Thermistor compensation methods for achieving transistor circuit stability. (2.5)

Ans. Refer to Q.1. (iii) (a) First Term Examination Sept. 2014.

Q.1. (c) Multistage Amplifier.

Ans. Refer to Q.1 (a) (iv) First Term Examination Sept. 2014.

Q.1. (d) Effect the negative feedback on the signal distortion of an amplifier.

Ans. Refer to Q.1. (ai) Second Term Examination, Nov. 2014.

Q.1. (e) CMOS Devices.

Ans. Refer to Q.1. (a ii) Second Term Examination, Nov. 2014.

Q.1. (f) Thyristors

Ans. Refer to Q.1. a(iii) Second Term Examination, Nov 2014.

Q.1. (g) Thermal stability of FETs

Ans. Refer to Q.1. a (iv) Second Term Examination, Nov. 2014.

Q.1. (h) Advantages of 'h'- parameters over other parameters.

Ans. Refer to Q.1. (a i) First Term Examination Sept 2014.

(II) Compare any five of the following:

(a) Hybrid π Model and the h -parameters Model for transistor analysis. (2.5×5 = 12.5)

Ans. h - Parameters model: For a small ac. signals a transistor behaves as a linear circuit. A linear circuit having input and output terminals can be analyzed by four parameters (ohm, mho and two dimension less) called hybrid parameters. The h -parameter model is typically suited to transistor circuit modeling. It is important because.

(i) Its values are used on specification sheets.

(ii) It is one model that may be used to analyze the circuit behaviour.

→ At low and mid-band frequencies, the h - parameter values are real values. This model is not suited for circuit analysis at high frequencies.

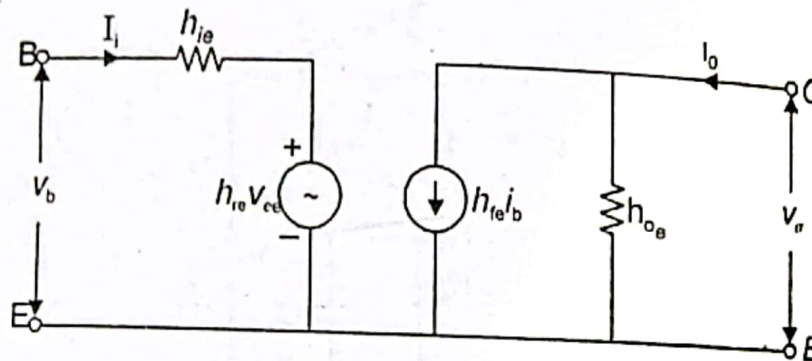


Fig. Hybrid equivalent circuit

The h parameter model is defined by

$$\begin{bmatrix} V_i \\ I_o \end{bmatrix} = \begin{bmatrix} h_{ie} & h_{re} \\ h_{fe} & h_{oe} \end{bmatrix} \begin{bmatrix} I_i \\ V_o \end{bmatrix}$$

$$h_{ie} \text{ (Input resistance)} = \frac{V_i}{I_i} \bigg|_{V_o=0} = \text{constant}$$

$$h_{re} \text{ (reverse voltage gain)} = \frac{V_i}{V_o} \bigg|_{I_i=0} = \text{constant}$$

$$h_{fe} \text{ (forward current gain)} = \frac{I_o}{I_i} \bigg|_{V_o=0} = \text{constant}$$

$$h_{oe} \text{ (output admittance)} = \frac{I_o}{V_o} \bigg|_{I_i=0} = \text{constant}$$

Note that lower case suffixes indicate small signal values and the last suffix indicates the mode.

Hybrid- π Model: It is a better model than h -parameter model since it contains frequency resistive components. The model can be quite accurate for low frequency circuits and can easily be adapted for higher frequency circuits with the addition of appropriate interelectrode capacitances and other parasitic elements. The full model

introduces the virtual terminal B', so that the base spreading resistance $r_{bb'}$ and $r_{be'}$ (resistance between B' and E) can be represented separately. $C_{b'e}$ is diffusion capacitance representing minority carrier storage in the base. The feedback components $r_{b'c}$ and $C_{b'c}$ are introduced to represent the early effect.

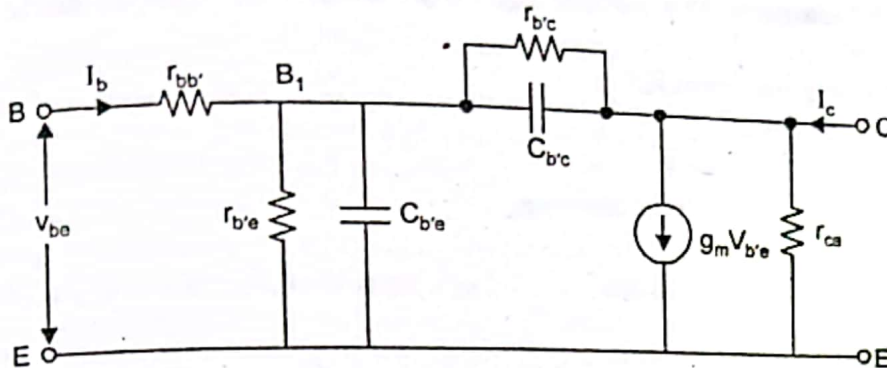


Fig: Hybrid- π Model at high frequencies

Q.2. (b) Cascaode and Cascode amplifiers:

Ans. Refer to Q.1. (b) (iii) First Term Examination Sept. 2014

Q.2. (c) Triac and Diac

Ans. Refer to Q.1 (bi) Second Term Examination Nov. 2014.

Q.2. (d) Shockley Diode and SCR.

Ans. Refer to Q.1. (b iii) Second Term Examination, Nov, 2014.

Q.1. (e) Positive and Negative feedback.

Ans. Refer to Q.1. (b iv) Second Term Examination Nov, 2014.

Q.1. (f) MOSFET and MESFET

Ans. MOSFET stands for Metal oxide semiconductor field effect transistor and MESFET stands for Metal semiconductor field effect transistor.

The main difference between MOSFET and MESFET is in the gate terminal fabrication. In MOSFET, between the gate and the channel, it is a thin layer of dielectric material SiO_2 and gate input behaves as a capacitor. In MESFET, between gate and the channel, there is just a metal semiconductor junction (such as happens in Schottky diode and JFET). The MOSFET's and MESFET's electrical equations are very similar. The main advantage of MESFET over the MOSFET is its speed (It has larger band width or transition frequency) but MESFET fabrication is more complicated. Often MESFET are build with Gallium Arsenide (GaAs) instead of silicon and GaAs crystals have in general many more defects than the Si used for all MOS processes.

MESFET's application is in microwave circuits, military applications and high performance communications.

Q.1. (g) Depletion and enhancement type MOSFET:

Ans. Refer to Q.6. (b), End Term Examination Sept 2014.

Q.1. (h) Emitter follower and Darlington Pair.

Ans. Refer to . 1 (b) (i). First Term Examination, Sept 2014.

UNIT-I

Q.2. (a) A Si transistor has fixed bias neglecting the leakage current I_{cBo} , if $V_{cc} = 15\text{ V}$, $R_B = 500\text{ K}\Omega$. Find I_C and V_{CE} for $\beta = 50$. What will be the effect on the working of transistor if $\beta = 101$?

Ans. Refer to Q.2 (a) First Term Examination Sept. 2014.

Q.2. (b) Sketch the gain-frequency response curve of a RC coupled amplifier, making the cut off frequencies and the bandwidth. Explain why the gain falls at high and low frequencies?

Ans. Refer to Q.2. (b) First Term Examination, Sept.. 2014. (4)

Q.2. (c) Compare the advantages and disadvantages of various biasing configurations. (3.5)

Ans. Types of bias circuit:

1. Fixed bias circuit
2. Collector to base bias
3. Fixed bias with emitter resistor
4. Voltage divider bias.

(1) Fixed bias: Advantages: (a) It is simple to shift the operating point anywhere in the active region.

(b) A very small number of components are required.

Disadvantages: (a) The collector current does not remain constant with variation in temperature or power supply therefore operating point is unstable.

(b) changes in V_{be} will change I_B and thus cause I_E to change. This in turn alters the gain of the stage.

(c) Operating point depends on β when transistor is replaced with another one.

→ Fixed bias is rarely used in linear circuit. It is often used in the circuit where transistor is used as a switch.

(2) Collector feedback bias: Advantage: Circuit stabilizes the operating point against variations in temperature and β

Disadvantages:

(a) Large changes in β will greatly change the operating point.

$$(b) \quad I_c = \beta I_b = \frac{\beta(V_{cc} - V_{be})}{R_b + R_c + \beta R_c} \approx \frac{V_{cc} - V_{be}}{R_c}$$

which is the case when $\beta R_c \gg R_b$.

Now, if R_c is large, a high V_{cc} is necessary which increases cost and if R_b is low, the reverse bias collector base region is small which limits the range of collector voltage swing that leaves the transistor in active mode. The R_b causes an A.C. feedback reducing the voltage gain of the amplifier.

(3) Fixed bias with emitter resistor:

Advantage: The circuit has the tendency to stabilize operating point against changes in temperature and β .

Disadvantage:

$$I_c = \beta I_b = \frac{\beta(V_{cc} - V_{be})}{R_b + (\beta + 1)R_E} \approx \frac{V_{cc} - V_{be}}{R_E}$$

Which is the case when

$$(\beta + 1)R_E \gg R_B$$

Now large value R_E causes large ac feedback which reduces the voltage gain of the amplifier.

(4) Voltage divider bias:

Advantage: (a) Only one dc supply is necessary (b) operating point almost independent of β and temperature

Disadvantage: To keep I_c independent of β , the following condition must be met.

$$I_c = \beta \frac{\frac{V_{cc}}{1 + R_1/R_2} - V_{be}}{(\beta + 1)R_E + R_1 \parallel R_2}$$

which is the case when

$$(\beta + 1)R_E \gg R_1 \parallel R_2$$

→ if R_E is of larger value, high V_{cc} is necessary

→ if $R_1 \parallel R_2$ is low draws more current from the power supply and lowers the input resistance of the amplifier.

Q.3. (a) A CE amplifier has a voltage source of internal resistance $r_s = 800 \Omega$ and the load impedance $R_L = 1000 \Omega$ the h parameters of the transistor are given as $h_{ie} = 1 \text{ k}\Omega$, $h_{re} = 2 \times 10^{-4}$, $h_{fe} = 50$ and $h_{oe} = 25 \mu\text{A/V}$. Compute the A_p , R_T , A_v and R_o . (5)

Ans. Refer to Q.3. (a) First Term Examination Sept 2014.

Q.3. (b) Draw and explain the circuit using diode compensation for the thermal changes in V_{BE} and I_{CO} ? (5)

Ans. Refer to Q.3. (b) First Term Examination Sept 2014.

Q.3. (c) What is a major drawback in the collector to base biasing method, and how is it avoided? (2.5)

Ans. Refer to Q.2. (c) First Term Examination Sept. 2014.

UNIT-II

Q.4. (a) A CE voltage amplifier with negative feedback produces a certain output voltage with an input of 100 mV. The same output voltage can be produced with only 10 mV. input if feedback is not used. Find out open loop voltage gain of amplifier if feedback factor $\beta = 0.05$. What is the output voltage (5)

Ans. Refer to 2 4(b), Second Term Examination, Nov. 2014.

Q.4. (b) Why overall gain of a multistage amplifier is less than that product of the gain of the individual stages? (2)

Ans. Refer to Q. 3(c) First Term Examination Sept 2014.

Q.4. (c) Describe a two stage R.C. coupled amplifier with a diagram, the advantages and disadvantages of R.C coupling and the differences in performance over a single stage amplifier. (5.5)

Ans. Two stage R.C coupled amplifier.

Mid Frequency Range: At this frequency the impedance offered by the coupling capacitor C_C is so small that it can be effectively considered as a short circuit and its effect be neglected.

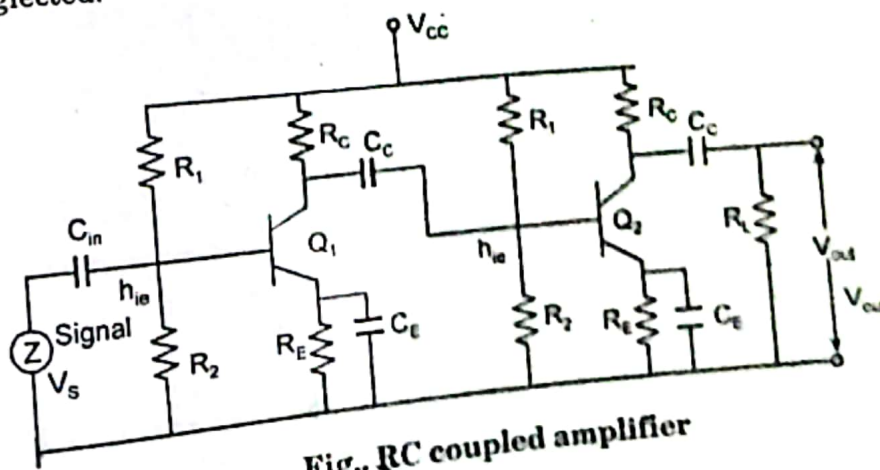


Fig.. RC coupled amplifier

$$V_{out} = h_{ie} I_c$$

$$V_{out} = \frac{h_{ie} h_{fe} I_b R_c}{R_c + h_{ie}}$$

$$V_{in} = h_{ie} I_b$$

$$\text{Voltage gain } A_{vm} = \frac{V_{out}}{V_{in}}$$

$$= \frac{h_{fe} R_c}{R_c + h_{ie}}$$

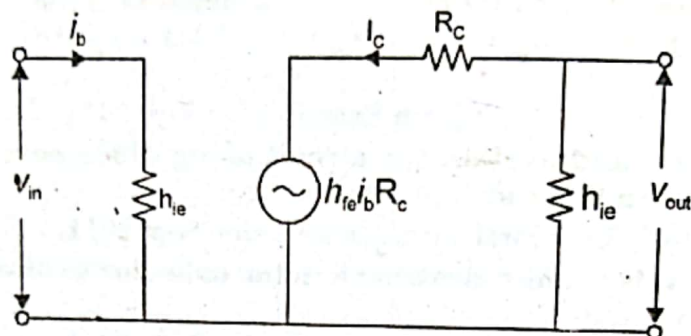


Fig. Equivalent Circuit for Mid Frequency Range

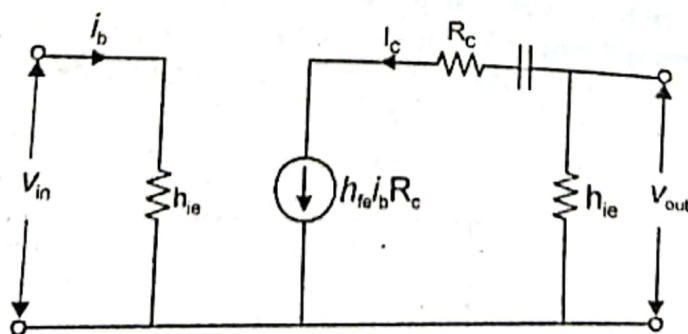
Low frequency Range: The impedance offered by C_c is comparable to R_c and 1 can not be neglected.

$$V_{out} = h_{ie} I_c = \frac{h_{fe} h_{ie} I_b R_c}{h_{ie} + R_c - \frac{j}{\omega C_c}}$$

$$V_{in} = h_{ie} I_b$$

$$\text{Voltage gain } A_{vl} = \frac{V_{out}}{V_{in}}$$

$$= \frac{h_{fe} R_c}{h_{ie} + R_c - j/\omega C_c}$$



Equivalent circuit for low frequency range

- Voltage gain decreases with decreases in frequency due to C_c

High frequency Range

- The reactance offered by C_c is very small & can be neglected.
- The inter-electrode capacitance C_{be} , C_{bc} & C_{ce} (replaced with C_d) offer a low impedance at input side this reduces the input impedance of the device and finally reduces the voltage gain at high frequencies.

$$V_{out} = I_c \times \frac{1}{j\omega C_d}$$

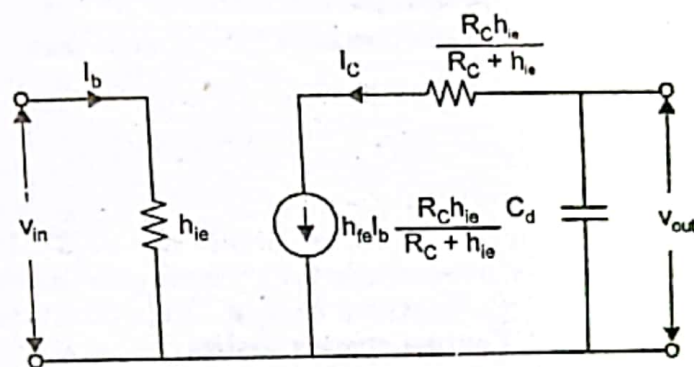
$$= \frac{h_{fe} h_{ie} I_b R_c}{h_{ie} R_c j\omega C_d + h_{ie} + R_c}$$

$$= h_{ie} I_b$$

∴ Voltage gain

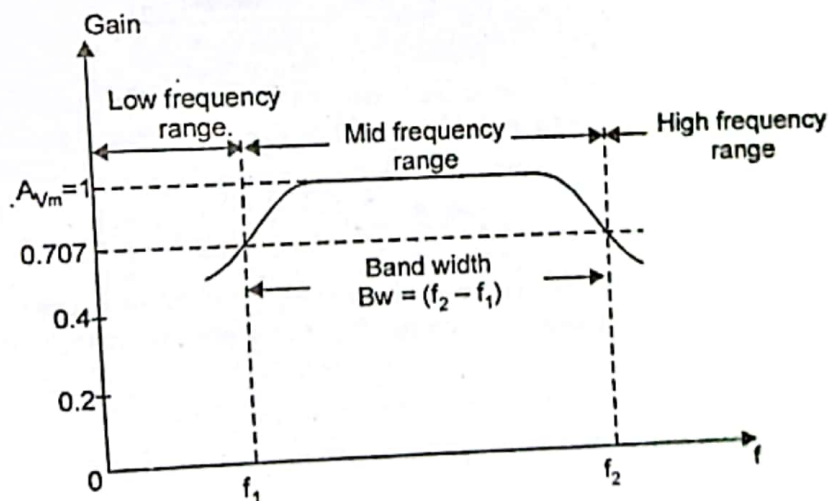
$$A_{vh} = \frac{V_{out}}{V_{in}}$$

$$A_{vh} = \frac{h_{fe} R_c}{h_{ie} R_c j\omega C_d + h_{ie} + R_c}$$



Equivalent Circuit for high frequency range

Frequency Response Curve



Advantages: (i) It has excellent frequency response. The gain is constant over the audio frequency range.

(ii) It has lower cost since it employs resistors and capacitors which are cheap.

(iii) The circuit is very compact as the modern resistors and capacitors are small and extremely light.

Disadvantages: (i) The RC coupled amplifiers have low voltage and power gain. It is because the low resistance presented by input of each stage to the preceding stage decreases the effective load resistance (R_{AC}) and hence the gain.

(ii) They have the tendency to become noisy with age.

(iii) Impedance matching is poor.

Q.5. (a) Discuss the various coupling schemes for coupling the stages of multistage amplifiers.

Ans. Refer to Q.4. (a) First Term Examination, Sept. 2014.

Q.5. (b) A CE amplifier has an open loop gain equal to 500 and the input resistance 1.0 kΩ. To raise the input resistance to 5.0 kΩ what type of feedback configuration and feedback factor β should be used?

Ans. Refer to Q.2. (c), Second Term Examination, Nov. 2014.

Q.5. (c) What is the effect of negative feedback on the bandwidth of an amplifier?
(2.5)

Ans. Refer to Q.4. (a) Second Term Examination, Nov. 2014.

UNIT III

Q.6. (a) Describe the important parameters of JFET.

Ans. The three important parameters of JFET are as follows:

- Dynamic Drain resistance (r_d)
- Transconductance (g_m)
- Amplification factor (μ).

(4)

Dynamic Drain Resistance (r_d):

Definition: Drain resistance is an AC resistance of a JFET. It is defined as the ratio of change in the drain to source voltage to the corresponding change in the drain current at a constant value of gate to source voltage. Drain resistance is calculated in the saturation region of the FET output characteristics.

$$\therefore r_d = \left. \frac{\Delta V_{DS}}{\Delta I_D} \right|_{\text{constant } V_{GS}} \quad \dots(1)$$

Amplification Factor (μ):

Definition: Amplification factor " μ " is defined as the ratio of change in the drain to source voltage, to change in the gate of source voltage, at a constant value of I_D .

$$\mu = \left. \frac{\Delta V_{DS}}{\Delta V_{GS}} \right|_{I_D \text{ constant}} \quad \dots(2)$$

Transconductance (g_m): The transconductance g_m is defined as the ratio of change in drain current to the corresponding change in gate to source voltage, at a constant value of drain to source voltage.

$$\therefore g_m = \left. \frac{\Delta I_D}{\Delta V_{GS}} \right|_{\text{constant } V_{DS}} \quad \dots(3)$$

$$M = \frac{\Delta V_{DS}}{\Delta I_D} \times \frac{\Delta I_D}{\Delta V_{GS}}$$

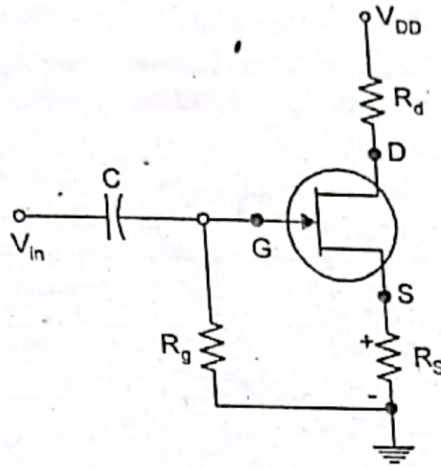
$$M = r_d \times g_m$$

As μ is the ratio of voltage it is unitless.

Q.6. (b) Describe the working of a UJT Relaxation Oscillator. How can it be used for triggering SCR?

Ans. Refer to Q.3. (a) Second Term Examination, Nov. 2014.

Q.6. (c) One n-channel FET amplifier (figure) has $V_p = -2.0V$, $g_{mo} = 1.6 \text{ mA/V}$ and $I_{DSS} = 1.65 \text{ mA}$. It is desired to bias the circuit at $I_D = 0.8 \text{ mA}$ using $V_{DD} = 24 \text{ V}$. Find (i) V_{GS} (ii) g_m (iii) R_s (4.5)



Ans.

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2$$

$$0.8 = 1.65 \left(1 + \frac{V_{GS}}{2} \right)^2$$

$$V_{GS} = -0.61 \text{ Volt}$$

$$g_m = g_{mo} \left(1 - \frac{V_{GS}}{V_P} \right)$$

$$g_m = 1.6 \left(1 - \frac{-0.61}{-2} \right) \text{ mA/V}$$

$$g_m = 1.6 (1 - 0.305) \text{ mA/V}$$

$$g_m = 1.6 \times 0.7 \text{ mA/V}$$

$$g_m = 1.12 \text{ mA/V}$$

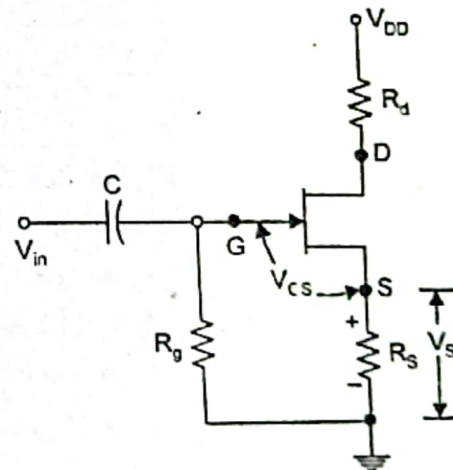
$$V_s = -V_{GS}$$

$$I_D R_s = 0.61$$

$$R_s = \frac{0.61}{0.8} \text{ k}\Omega$$

$$R_s = 0.763 \text{ k}\Omega$$

$$R_s = 763 \Omega$$

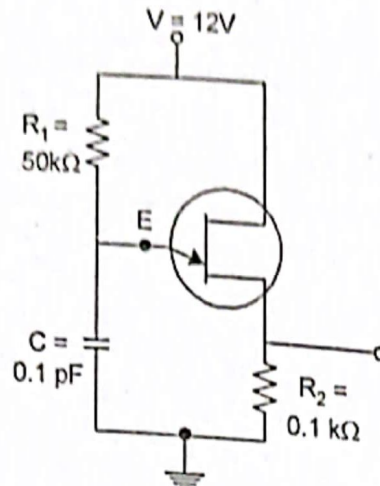


Q.7. (a) What are the precautions observed in handling MOSFET components? (2.5)

Ans. MOSFET has an ultra-thin silicon dioxide layer between the channel and gate. Because the insulating layer is so thin, it is easily destroyed by excessive gate-source voltage V_{GS} application of large gate voltage, an open-circuit gate may accumulate enough charge so as to develop an electric field large enough to puncture the thin SiO_2 .

layer. SiO_2 may also destroyed if MOSFET is insulated or removed from a circuit while the power is ON. Transient voltages caused by inductive pickback and other effects may exceed $V_{GS(\text{max})}$ rating this will wipe out the MOSFET. Generally ground rings are used to short all leads of a MOSFET for avoiding any voltage build up between gate and source. The grounding rings are removed after MOSFET is connected in the circuit.

Q.7. (b) Given the Relaxation Oscillator in the figure, find (i) values of R_{B1} and R_{B2} at $I_E = 0 \text{ A}$, (ii) Calculate V_P , the voltage required to trigger the UJT 'ON'. (5)



Ans.

$$R_{BB} = 5 \text{ k}\Omega$$

$$\eta = 0.6$$

$$V_V = 1 \text{ V}$$

$$I_V = 10 \text{ mA}$$

$$I_P = 10 \mu\text{A}$$

$$R_{B1} = R_{B2} = R_{BB}$$

$$\eta = \frac{R_{B1}}{R_{BB}}$$

$$R_{B1} = \eta R_{BB} = 0.6 \times 5 \text{ k}$$

$$R_{B1} = 3 \text{ k}\Omega$$

$$R_{B2} = R_{BB} - R_{B1} \\ = (5 - 3) \text{ k}\Omega$$

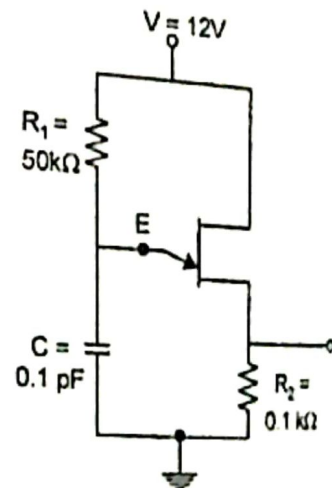
$$R_{B2} = 2 \text{ k}\Omega$$

$$V_P = \eta V_{BB} + V_B$$

$$V_P = 0.6 \times 12 + 0.7 \text{ (} V_B = 0.7 \text{ for silicon)}$$

$$V_P = 7.2 + 0.7$$

$$V_P = 7.9 \text{ volt.}$$



Q.7. (c) Describe some important applications using SCRs

Ans. Refer to Q.2. (b) Second Term Examination, Nov. 2014

UNIT-IV

Q.8. (a) Describe various classifications of Power amplifiers based on their
 Q.8. (a) Second Term Examination, Nov. 2014. (4)

Q.8. (b) What is the advantage of using a class B amplifier as compared to a
 Class A amplifier? (4)

Ans. Class B Power Amplifiers

In class B operation the transistor is so biased that zero signal collector current is
 Hence class B operation does not need. Any biasing system. The operating point is
 at cutoff, as illustrated in Fig.. It remains forward biased for only half cycle of the
 input signal i.e., its conduction angle is 180° .

As illustrated in Fig., during the positive half cycle of the input ac signal, the circuit
 is forward biased and, therefore, collector current flows. On the other hand, during
 negative half cycle of the input ac signal, the circuit is reverse biased and no collector
 current flows.

Power and Efficiency Calculations

Input dc power, $P_{in} (dc) = V_{cc} I_{dc}$ where I_{dc} is the average or direct current taken from
 the collector supply.

If $I_{c max}$ is the maximum or peak value of collector or output current, then

$$I_{dc} = \frac{1}{2\pi} \int_0^\pi I_{c max} \sin \theta d\theta = \frac{I_{c max}}{2\pi} [-\cos \theta]_0^\pi = \frac{I_{c max}}{\pi}$$

$$\therefore i_c = I_{c max} \sin \theta \text{ for } 0 < \theta < \pi.$$

$$\text{RMS value of collector current, } I_{c rms} = \frac{I_{c max}}{\sqrt{2}}$$

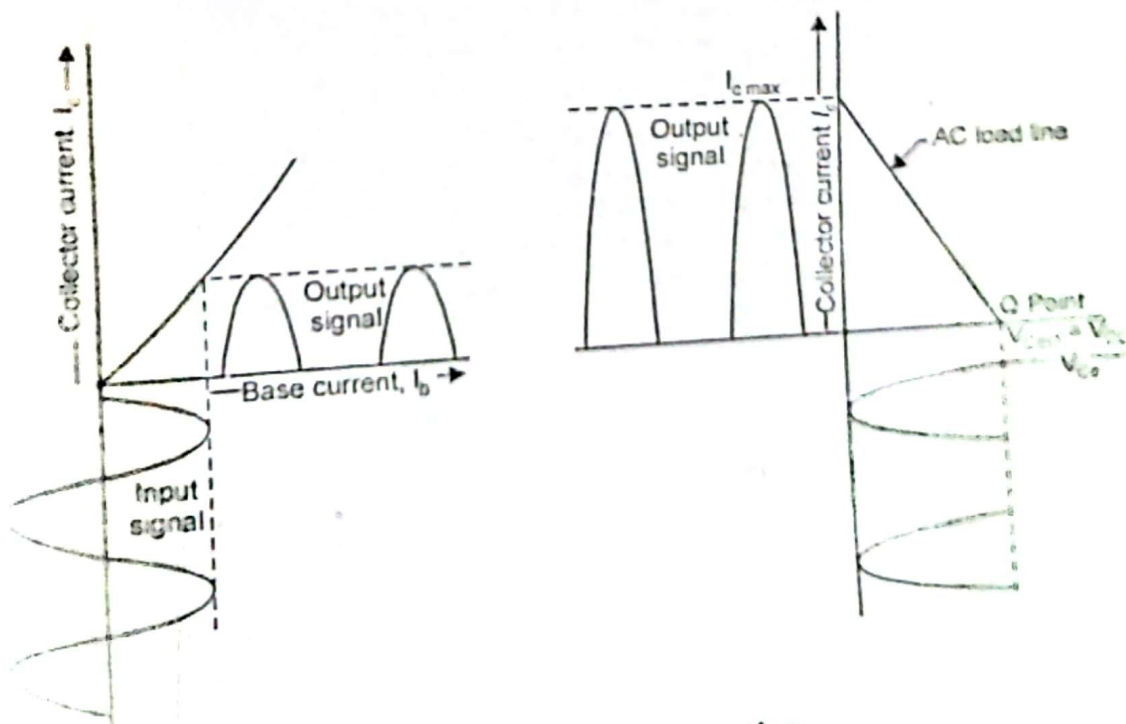


Fig. Class B Operation

RMS value of output voltage, $V_{rms} = \frac{V_{CC}}{\sqrt{2}}$

Hence output power during half cycle,

$$P_{out(ac)} = \frac{1}{2} \frac{I_{c max}}{\sqrt{2}} \times \frac{V_{CC}}{\sqrt{2}} = \frac{I_{c max} V_{CC}}{4}$$

In above equation factor $\frac{1}{2}$ is used because power is developed during one half cycle only.

$$DC \text{ power loss in load, } P_{R_{C(dc)}} = I_{dc}^2 R_C = \left(\frac{I_{c max}}{\pi} \right)^2 R_C$$

DC power loss in collector region or transistor

$$= P_{in(dc)} - P_{R_{C(dc)}} - P_{out(ac)}$$

Overall efficiency,

$$\eta_{overall} = \frac{P_{out(ac)}}{P_{in(dc)}} = \frac{I_{c max} V_{CC} / 4}{I_{c max} V_{CC} / \pi}$$

$$= \frac{\pi}{4} = 0.785 \text{ or } 78.5\%$$

Conclusions: 1. Severe distortion occurs because of total absence of negative half cycle from the output.

2. As average current in class B operation $\frac{(I_{c max})}{\pi}$ is less than that in class A operation, so power dissipation is less and consequently overall efficiency is high [78.5% when peak input signal makes $V_{ce min} = 0$]. In general, overall efficiency is given as

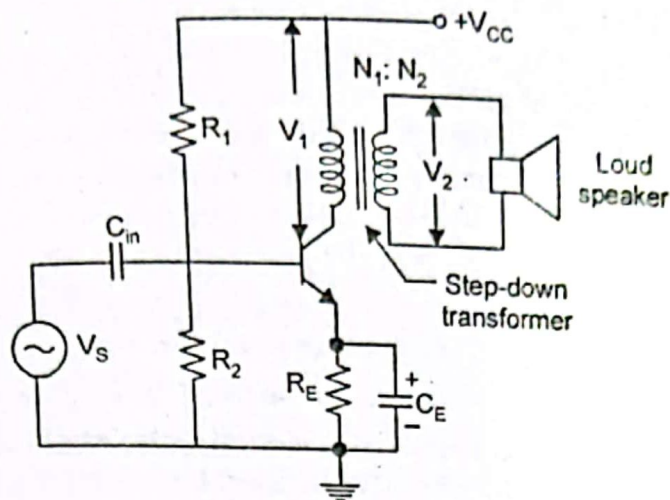
$$\eta_{overall} = 78.5 \left[1 - \frac{V_{ce min}}{V_{CC}} \right] \%$$

3. Zero signal input represents the best condition for class B operation as collector current is zero. This is quite contrary to class A operation. In class B operation the transistor dissipation increases with creased input signals.

Q.8. (c) Describe a transformer-coupled class A audio amplifier. What is the advantages of transfoer coupling? (4.5)

Ans. This is also sometimes referred to as single ended power amplifier.

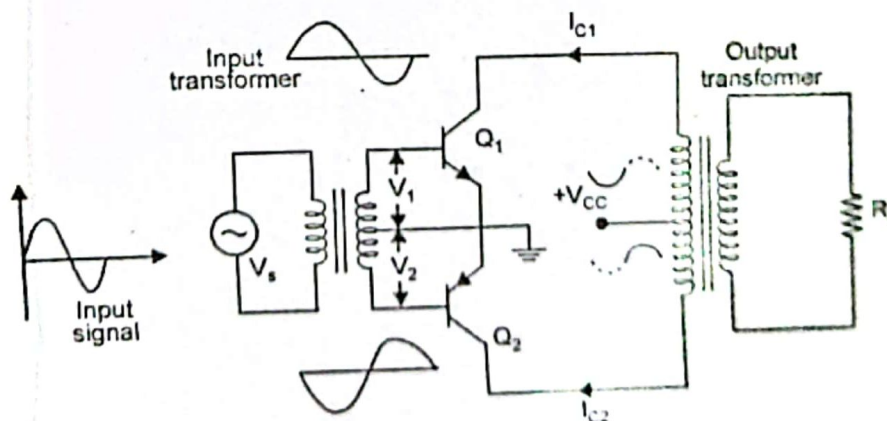
In case of direct coupled (without transformer) class A power amplifier, the quiescent current flows through the collector resistive load causes large wastage of dc power in it. This dc power dissipated in the load resistor does not contribute to the useful ac output however. Further more it is inadvisable to pass the dc through the output device such as in a voice coil of a loud speaker. For these reason an arrangement using a suitable transformer for coupling the load (loud speaker) to the amplifier is usualluy employed as shown in figure. This arrangement enhances the conversion efficiency by a factor of 2 by eliminating the dc power dissipation in the load. Since the primary of the transformer has negligible dc resistance. The ac power is, however coupled magnetically across the transformer into the load R_L . This coupling method also prevents a large dc current



from flowing through the load which otherwise could be harmful if the load were a loudspeaker. This arrangement also permits impedance matching. A step-down transformer of suitable turn ratio $\left(\alpha = \frac{N_1}{N_2}\right)$ is provided to couple the high impedance collector circuit to low impedance load.

Q.9. (a) Describe with a diagram operation of a class B push pull amplifier. (5)

Ans. Class B push-pull operation is the same as that for the class A operation except that the devices are biased at cut-off. It gives higher operating efficiency theoretical 78.5% due to the fact that no power is drawn by the circuit under zero-signal condition.



Circuit operation: When the input signal is applied the centre tapped secondary of the input transformer develops two signals which are identical but in phase opposition. The transistors Q_1 and Q_2 are driven by these two signals. Thus when V_1 is going positive, V_2 is going negative, so that transistor Q_2 is being biased further off when transistor Q_1 is being biased ON. As the collector current in Q_1 increases from zero, it produces a half wave of voltage across upper half of the primary of the output transformer. When the positive half cycle of the input signal to Q_1 begins to go negative, the signal at Q_2 base is commencing to go positive. Thus Q_1 becomes biased off again, Q_2 is biased ON

and a half cycle of voltage waveform is generated across the lower primary winding of output transformer.

The effect of the two-half cycles in separate halves of the primary of the output transformer is to produce a magnetic flux in the transformer core, which flows first in one direction and then in the opposite direction. This flux links with the secondary of the output transformer and generates a complete sine-wave output, which is passed on to the load. Thus, in the class B circuit, the two transistors are said to operate in push-pull.

Q.9. (b) Describe 'cross over' distortion in amplifiers.

Ans. Refer to Q.9. (b) End Term Examination, Sept. 2014. (3.5)

Q.9. (c) Calculate the transformer turn ratio required to match a 8Ω speaker load to an amplifier so that the effective load resistance is $7.2\text{ K}\Omega$. (4)

Ans.

$$R'_L = \alpha^2 R_L$$

where R'_L is effective resistance looking into transformer primary R_L is load connected across the transformer secondary α is ratio of primary to secondary turns of step down transformer

$$\therefore \alpha = \frac{N_1}{N_2}$$

$$\text{and} \quad \left(\frac{N_1}{N_2}\right)^2 = \frac{R'_L}{R_L} = \frac{7.2\text{ k}\Omega}{8\pi} = \frac{7200}{8}$$

$$\frac{N_1}{N_2} = \sqrt{\frac{900}{1}}$$

$$\frac{N_1}{N_2} = \frac{30}{1}$$

$$\therefore N_1:N_2 = 30:1$$